The Soma-Field: A Wave-Based Model of Emotional Dynamics and Its Clinical Implications

Bridging Quantum Field Theory, Neural Energy Functions, and Somatic Psychotherapy

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Abstract

Since McCulloch and Pitts (1943), artificial neural networks have provided increasingly sophisticated formal models of one component of biological intelligence: the neocortex — pattern recognition, sequence prediction, and error minimisation. The complementary component — the limbic system, responsible for valuation, threat detection, arousal modulation, and the somatic state reinstatement that underlies trauma — has received no formal mathematical treatment. This paper proposes the **Soma-Field Model**: the first formal field-theoretic architecture for the limbic system and its coupling to the body and autonomic nervous system.

Drawing on the energy-function formalism of Hopfield neural networks and a formal identification with Quantum Field Theory — not as metaphor but as shared mathematical structure — the model provides a mathematically grounded account of four clinically fundamental phenomena: the sub-perceptual persistence of emotional states; the threshold at which emotion enters conscious awareness; the attractor structure corresponding to fight, flight, freeze, and regulated calm; and the formal mechanism of complex PTSD as a specific configuration of the attractor landscape. The conscious emotional percept is identified as the one-dimensional impulse response (Green's function) of an eleven-dimensional coupling manifold, placing emotional experience and fundamental particles in the same mathematical category: both are poles in the propagator of their respective field.

The soma-field adds the component that has been absent from every artificial neural network since McCulloch and Pitts (1943): a formal model of the limbic system. Together, the Hopfield framework and the Soma-Field Model constitute the first complete formal description of the two principal computational substrates of the vertebrate brain. The body schema and somatic pain states are incorporated as field modes, grounding the model in interoceptive neuroscience and phantom limb research. A companion instrument for therapeutic use and a formal abstract film specification are described as practical applications. Clinical implications for assessment, psychoeducation, and the treatment of complex trauma are discussed. A companion experiment (QUANT-EXP-1, May 2026) demonstrates that the quantum extension of this model — transverse-field quantum annealing on the same Hopfield landscape — traverses topological barriers that the tested low-noise classical dynamics does not cross, providing bounded empirical support for the therapeutic mechanism theorem in this model class.

Contents

AI has had a brain since 1943. Now it has a body.

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1 1. Introduction

A patient sits with their therapist and is asked: *"What are you feeling right now?"* The question is deceptively simple. They may say *anxious*, yet that word covers a vast and heterogeneous territory — a tightness in the chest, a running commentary of worry, a vague readiness to flee, a memory surfacing from childhood. Another patient, asked the same question, reports feeling nothing at all; and yet their posture, respiration, and the quality of their silence suggest otherwise. The emotion is there. It is simply not yet conscious.

This gap between emotional presence and emotional awareness is one of the most clinically significant phenomena in psychotherapy. Theories of affect regulation (Schore, 2001), somatic experiencing (Levine, 2010), sensorimotor psychotherapy (Ogden, Minton & Pain, 2006), and polyvagal theory (Porges, 2011) all grapple, in different ways, with the same observation: emotions exist in the body before — and often without —

being named in the mind. Eugene Gendlin called the sub-verbal bodily sense of an emotional situation the *felt sense* (Gendlin, 1978): something that is there, whole and present, but not yet articulate.

The Soma-Field Model proposed here attempts to give this clinical observation a formal structure. It does so by borrowing a conceptual tool from physics: the field. In physics, a field is not a thing that exists at a point. It is a quantity that exists everywhere in a space, continuously, whether or not it is observed. Particles — the things we can measure — are not separate from the field; they are *excitations* of it, local concentrations of energy that arise when the field is perturbed above a certain threshold.

The central claim of this paper is that this structure accurately describes the phenomenology of emotion. The emotional field is always there, distributed across body and nervous system. What we call a conscious emotional experience is an excitation of that field — a local concentration that has crossed a perceptual threshold and entered awareness. The field continues below the threshold whether or not we attend to it, and its sub-perceptual activity shapes our behaviour, physiology, and cognition continuously.

The Soma-Field Model contributes the first formal field-theoretic architecture for the limbic system. Every artificial neural network since McCulloch and Pitts (1943) [@mcculloch1943] is a formal model of the neocortex — the pattern-recognition and prediction layer. The limbic system — responsible for emotional valuation, threat detection, and the somatic state reinstatement that underlies trauma — has never received a comparable formal treatment. The Soma-Field Model is that treatment. Together with the Hopfield framework, it constitutes the first complete formal description of the two principal computational substrates of the vertebrate brain.

The paper proceeds as follows. Section 2 reviews the relevant background in somatic clinical models, and introduces the two theoretical tools borrowed from physics and computer science: quantum field theory and Hopfield network energy functions. Section 3 develops the Soma-Field Model in detail. Section 4 describes the energy landscape, including the attractor states corresponding to fight, flight, freeze, and regulated calm. Section 5 discusses dissonance and resolution as mechanisms of emotional interaction. Section 6 describes the Soma-Field Instrument, a practical tool for therapeutic use. Section 7 addresses clinical implications.

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2 2. Background

2.1 2.1 The Body-Mind Problem in Clinical Practice

Contemporary neuroscience has largely dissolved the Cartesian boundary between body and mind. Damasio (1994) demonstrated that emotion is inseparable from rational cognition: patients with damage to the ventromedial prefrontal cortex — preventing the normal generation of somatic signals — lose not only their emotional range but also their capacity for effective decision-making. Van der Kolk (2014) documented extensively how traumatic emotional states are encoded not merely in explicit memory but in posture, gesture, visceral sensation, and autonomic regulation. Porges' polyvagal theory (2011) provided a neurobiological account of how the autonomic nervous system generates three hierarchically organised states — ventral vagal (social engagement), sympathetic (mobilisation: fight/flight), and dorsal vagal (immobilisation: freeze) — each with characteristic phenomenological and behavioural signatures.

What these frameworks share is a conviction that emotional states are not located in the brain alone, nor in the body alone, but in a coupled system that is best understood as a single functional unit. The term *soma* — from the Greek for body — is used here to denote this unified body-mind system, following the tradition of somatic psychotherapy.

2.2 2.2 The Felt Sense and Sub-Perceptual Emotion

Gendlin's concept of the *felt sense* (1978) is of particular relevance. He described it as "a special kind of internal bodily awareness… a body sense of meaning." It is not an emotion in the ordinary sense — not a named feeling — but something more diffuse: a pre-articulate sense that *something is there*, present in the body, before it has been identified or named. Focussing, the therapeutic method Gendlin developed, works precisely by attending to this pre-threshold signal and allowing it to surface into conscious articulation.

The Soma-Field Model provides a formal account of what the felt sense is: it is the activity of the emotional field below the perceptual threshold. It is real, causal, and continuously present. It shapes cognition and behaviour even when it does not surface as a named feeling.

2.3 2.3 Quantum Field Theory: Structure, Not Metaphor

Quantum Field Theory (QFT) is the framework of modern particle physics. Its central departure from classical physics is the priority of the *field* over the *particle*. In QFT, what we call particles — electrons, photons — are not fundamental objects. They are *excitations* of an underlying field: local, stable configurations of energy that arise when the field receives a sufficient perturbation.

The quantum vacuum — the ground state of the field — is not empty. It is a seething background of virtual fluctuations: momentary excitations that do not have enough energy to persist as observable particles. The vacuum is active, but sub-threshold.

A SINGLE FIELD MODE — amplitude over time

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(e.g. a mode of the electromagnetic field; or, later, a mode of the emotional field)

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└─────────────────────────────────────────────────────────────► time

←── VIRTUAL: field fluctuates but stays sub-threshold ──────────→ ←REAL→
     present, active, causally real — but not locally detectable       ↑
    (the QUANTUM VACUUM: not empty; seething with activity)       particle
                                                                   created
```

Figure 0. A single field mode in quantum field theory. The field oscillates continuously. Below the detection threshold T, excitations are sub-threshold — real and causally active, but not detectable as particles. The quantum vacuum is not empty; it is a field in constant motion that never quite crosses the threshold. When the amplitude does cross T, a particle exists: a locally observable excitation. The same structure — field always present, consciousness only when threshold crossed — is the core of the Soma-Field Model.

This paper does not claim that emotions are quantum phenomena in any literal sense: the soma-field is a classical field, not a quantised one. The claim is stronger and more specific than analogy: the mathematical object being constructed — the Green's function of a coupled field manifold — is formally of the same *type* as the objects that arise in QFT, differing only in the dimensionality of the manifold and the nature of the probe. What was previously described as a structural analogy is here identified as a formal correspondence: a particle is a pole in the propagator of its field; a conscious emotional percept is a pole in the propagator of the soma-field. Different physics. Same mathematics.

That correspondence gives the model precise vocabulary for the following set of ideas, which are central to the clinical observation of emotion:

- A quantity that exists everywhere, continuously, even when unobserved
- A background of sub-threshold activity that is real and causally effective
- The emergence of observable phenomena (conscious feelings) through threshold-crossing excitation of that background
- The possibility of multiple simultaneous excitations that interact with one another

Note (May 2026): A subsequent experiment (QUANT-EXP-1) demonstrates that the quantum extension of the Hopfield landscape used in this model — replacing the classical Langevin process with a transverse-field quantum annealer — produces a measurable *topological reachability advantage*: quantum annealing reaches attractor basins that cold classical dynamics cannot reach at any finite noise level. This upgrades the formal correspondence from a structural claim to a testable empirical prediction. See `paper/QUANT-EXP-SWEEP-2026-05-20.md` and `paper/quantum-soma-penrose.md` for the full results and theoretical implications.

One further consequence follows. The clinical phenomena of alexithymia — difficulty identifying and naming feelings — and its apparent opposite, emotional flooding or hypervigilance, have always been treated as separate conditions requiring separate explanations. In the Green's function framing, they are the same structure at two extremes of the same parameter: the perception threshold T_i is too high (the bulk dynamics cannot cross into observable experience) or too low (bulk fluctuations flood the boundary without filtering). This is structurally identical to one of the deepest open problems in particle physics — the **hierarchy problem** — which asks why gravity is so much weaker than the other forces. The standard answer is that gravity propagates in the full higher-dimensional bulk while other forces are confined to a lower-dimensional brane; the coupling across the brane boundary determines the apparent weakness. The soma-field correspondence is exact: the threshold T_i *is* the brane. Perception is confined to the one-dimensional boundary of an eleven-dimensional dynamics. The hierarchy of emotional experience — why conscious feeling is so much weaker and more transient than the underlying field activity — has the same formal structure as the hierarchy of forces.

2.4 2.4 Neural Network Energy Functions and Hopfield Networks

In 1982, John Hopfield (awarded the Nobel Prize in Physics in 2024) proposed a model of associative memory based on a network of interconnected neurons (Hopfield, 1982). The critical insight was borrowed directly from statistical physics: the network could be assigned an **energy function** — a scalar quantity that decreases with each state update — such that the network would always evolve toward a local energy minimum. These minima are the stable states of the network: its memories, or more precisely, its *attractors*.

Hopfield observed that his neural network's dynamics were mathematically identical to those of an Ising spin-glass model from condensed matter physics — a system of interacting magnetic spins that minimises its total energy by aligning or anti-aligning with neighbours. The energy function he used is:

$$H(\mathbf{s}) = -\frac{1}{2}\sum_{i,j} W_{ij}\, s_i s_j - \sum_i \theta_i s_i$$

where $\mathbf{s}$ is the state of the network, W_{ij} is the coupling strength between units i and j, and θ_i is the activation threshold of unit i. The network always moves in the direction of decreasing H.

The Soma-Field Model applies this energy function directly to emotional dynamics. The *emotional coupling matrix* W encodes the relationships between emotional modes — which emotions amplify one another, which suppress one another — and the energy function determines the direction in which the emotional field naturally evolves.

Hopfield's network is a formal model of the *neocortex*: a system for storing cognitive patterns and retrieving them from partial cues by minimising an energy function. Every artificial neural network constructed since McCulloch and Pitts (1943) [@mcculloch1943] — from perceptrons to backpropagation networks to transformers — sits in this neocortical lineage. These systems recognise patterns, predict sequences, and minimise prediction error with increasing sophistication. None of them possess a limbic system. They have no internal valuation, no arousal modulation, no threat-detection architecture, no attachment structure, no interoception. They have very effective cortex.

The Soma-Field Model does not add to the neocortical lineage. It proposes the architectural layer that has never been formally built: *an artificial limbic system.*

Hopfield memory is associative and pattern-completing; somatic memory is state-reinstating. The field does not merely remember what happened. It re-lives it. *A body with a past.*

Hopfield's later-reported wish to have incorporated something analogous to 'maternal instincts' into the energy function was, in this reading, not a desire for a better cortex. It was an intuition pointing directly at the absent system — the layer beneath the cortex that assigns value, registers threat, and holds the body in a particular way of being long after the event that caused it.

This positions the Soma-Field Model not as a supplement to the neocortical lineage but as its completion. Artificial neural networks have, for eighty years, been increasingly sophisticated formal models of the neocortex: pattern recognition, sequence prediction, error minimisation. The cortex has been mapped in extraordinary detail. The limbic system — which assigns value, detects threat, modulates arousal, maintains attachment, and reinstates whole somatic states in response to partial cues — has had no comparable formal treatment. The architectural description of the vertebrate brain was, until this paper, half-built.

Four kinds of formal intelligence. This architectural gap can be situated within a wider taxonomy. Four quotients have been proposed to describe the landscape of biological intelligence across popular and scientific usage. They map onto the formal components of this model with an exactness that is not coincidental:

Quotient	What it measures	Biological substrate	Soma-Field status
IQ — cognitive	Pattern recognition, reasoning, prediction	Neocortex	Built (1943–): McCulloch & Pitts $\to$ Hopfield $\to$ transformers
EQ — emotional	Valuation, arousal, affect regulation	Limbic system	**Built here**: W, $K(\tau)$, $H(\mathbf{e})$, C_{HRV}, $\dot{H}$

Quotient	What it measures	Biological substrate	Soma-Field status
AQ — adversity	Structural resilience under threat	PFC–limbic axis	**Built here**: S_{inst}, $\partial\|W\|/\partial t$, $C_{\text{HRV}}^{\text{recovery}}$
SQ — social	Attunement, theory of mind, relational navigation	Mirror system, TPJ	*Next paper*: κ_r, multi-field coupling

Table 3. Four dimensions of biological intelligence mapped onto the Soma-Field Model. The neocortical lineage (IQ) has been formally modelled for eighty years. Emotional intelligence (EQ) and adversity resilience (AQ) are formalised here for the first time. Social intelligence (SQ) is defined as the next extension of the framework.

AQ — adversity quotient — is formally the capacity to update W after adversity without the adversity permanently becoming W. Its mathematical definition appears in Section 3.4; its pathological lower bound is C-PTSD, in which all three components of AQ are simultaneously compromised (Appendix B.2).

The AI alignment implication follows directly. Current artificial systems have high IQ by construction and zero EQ, AQ, or SQ. The absence of internal valuation means that valuation must be injected externally — through reinforcement learning from human feedback (RLHF) and related techniques — which is structurally brittle for the same reason that a field with no limbic layer is brittle: the system has no internal stake in what it does. The Soma-Field formalisation specifies what that internal stake would look like, were it ever built.

A further lineage note is worth recording. Ramsauer et al. (2020) demonstrated that continuous-state modern Hopfield networks are mathematically equivalent to the self-attention mechanism in transformer language models. The softmax attention operation that drives contemporary large language models is a Hopfield retrieval step. The Soma-Field Model sits in this same energy-based lineage: the equations underlying associative memory, language understanding, and somatic trauma response are, at the appropriate level of abstraction, the same equations.

A historical irony completes the picture. String theory was not discovered as a theory of strings. In 1968, Gabriele Veneziano wrote down a scattering amplitude — a response function encoding how particles scatter — and only later did Nambu, Nielsen, and Susskind identify the string as whatever object produces that amplitude [@veneziano1968]. The response function came before the thing. The Soma-Field Model recapitulates this historical order deliberately: the primary object is the eleven-dimensional coupling manifold; the string — the one-dimensional conscious percept — is what the manifold produces when probed. We retain Veneziano's discovery and decline to reify the string.

2.5 2.5 The Formal Correspondences: Where the Link Was Seen

The structural analogy between QFT and the Soma-Field Model is not merely conceptual. There are three places where equations from different disciplines become, after substituting the relevant quantities, literally the same functional form. The following sets them side by side. The point is not to impress with notation but to show exactly where the recognition happened — the moment when the same Greek letters appeared in the same positions in two fields that had no prior reason to be connected.

The same Hamiltonian: Ising spin model (condensed matter physics, 1920s) — Hopfield neural network (computational neuroscience, 1982) — Soma-Field Model:

$$H_{\text{Ising}}(\sigma) = -\frac{1}{2}\sum_{i,j} J_{ij}\,\sigma_i\,\sigma_j - \sum_i h_i\,\sigma_i$$

$$H_{\text{soma}}(\mathbf{e}) = -\frac{1}{2}\sum_{i,j} W_{ij}\,e_i\,e_j - \sum_i \theta_i\,e_i$$

Replace $J_{ij} \to W_{ij}$, $\sigma_i \to e_i$, $h_i \to \theta_i$: identical. The physicist, the neural network theorist, and the somatic clinician are computing the same energy function on different state spaces. The Hopfield 2024 Nobel Prize was awarded for discovering this identity between spin physics and neural computation; the Soma-Field Model extends the same identity one step further to emotional dynamics.

The Wick rotation — why the same exponential appears in QM and in memory:

In quantum mechanics, the time evolution operator is a complex phase:

$$U(t) = e^{-i\hat{H}t/\hbar}$$

Substitute $t \to -i\tau$ (the *Wick rotation* — replacing real time with imaginary time):

$$e^{-i\hat{H}(-i\tau)/\hbar} = e^{-\hat{H}\tau/\hbar}$$

The oscillating complex exponential becomes a real decaying exponential. This is the Boltzmann weight $e^{-\beta\hat{H}}$ at $\beta = \tau/\hbar$. The Langevin equation $\dot{\mathbf{e}} = -\nabla H + \eta$ is the classical limit of this Wick-rotated dynamics. Every simulation of the soma-field running this equation is, formally, a path integral in imaginary time.

The same propagator: Euclidean QFT (imaginary-time two-point correlator for a massive scalar field) — C-PTSD trauma memory kernel:

$$G_E(\tau) = \langle \phi(0)\, \phi(\tau) \rangle_{\mathrm{QFT}} = \frac{1}{2m}\, e^{-m|\tau|}$$

$$K_{\mathrm{trauma}}(\tau) = \sum_k A_k\, e^{-|\tau|/\tau_k}$$

Same form. The QFT field mass m corresponds to $1/\tau_k$ — the reciprocal of the trauma trace decay time. A heavier particle has a shorter-range propagator; a shorter-lived trauma trace decays faster. Therapeutic processing (reducing A_k, increasing τ_k) is, in the QFT language, changing the mass and amplitude of the propagator until the correlation function vanishes.

The specific visual moment: the quantum phase factor is $e^{-i\omega t}$. Remove the i (Wick rotation) and it becomes $e^{-\omega\tau}$. The memory kernel is $e^{-\tau/\tau_k}$. These are the same exponential. The i is the only difference between a quantum field that oscillates and a trauma trace that decays.

QFT quantity	Symbol	Soma-Field analogue	Symbol
Field mode	ϕ_k	Emotional mode	e_i
Coupling constant	J_{ij}	Coupling matrix entry	W_{ij}
Field mass	m	Inverse decay time	$1/\tau_k$
Propagator amplitude	$1/2m$	Trauma trace amplitude	A_k
Euclidean propagator	$G_E(\tau) \propto e^{-m\tau}$	Memory kernel	$K(\tau) \propto e^{-\tau/\tau_k}$

QFT quantity	Symbol	Soma-Field analogue	Symbol
Vacuum energy	$\langle H \rangle_0$	Resting field energy	$H(\mathbf{e}_{\text{calm}})$
Thermal fluctuation	$k_B T$	Noise amplitude	σ_0
Wick rotation	$t \to -i\tau$	Real-time Langevin	$\dot{\mathbf{e}} = -\nabla H + \eta$

Table 2. Formal correspondence between QFT quantities and Soma-Field analogues. Each row is a single mathematical entity in two notations. These correspondences were not constructed after the fact; they are the reason the QFT framework was recognised as relevant.

The central identification — particle and percept as poles in their respective propagators. All four correspondences above follow from one structural fact. In QFT, a particle is not a separate object from the field. It is a *pole* in the field's propagator — the Green's function evaluated in momentum space:

$$\tilde{G}_{\text{QFT}}(k^\mu) = \frac{i}{k^2 - m^2 + i\varepsilon}$$

The particle exists precisely when the four-momentum satisfies $k^2 = m^2$ — the *on-shell condition*. The particle is the singularity in the field's response to a point source: the field's Green's function, evaluated at its own resonance.

Diagonalise W with eigenvalues λ_i (the natural resonance frequencies of the emotional modes). The soma-field propagator — the two-point correlator $\langle e_i(t)\, e_i(t') \rangle$ in the frequency domain — is:

$$\tilde{G}_{ii}(\omega) = \frac{\sigma_{\text{eff}}^2}{\omega^2 + \lambda_i^2}$$

A conscious emotional percept in mode i exists precisely when the excitation frequency ω approaches $i\lambda_i$ — the mode's natural resonance. The percept is the singularity in the soma-field's response to a somatic probe.

Setting the two propagators side by side:

$$\underbrace{\frac{i}{k^2 - m^2 + i\varepsilon}}_{\text{QFT: particle at mass-shell } k^2=m^2} \qquad \longleftrightarrow \qquad \underbrace{\frac{\sigma_{\text{eff}}^2}{\omega^2 + \lambda_i^2}}_{\text{Soma-Field: percept at resonance } \omega=i\lambda_i}$$

Both are poles in the propagator of their respective field manifold. A photon is not the electromagnetic field; it is the field's Green's function evaluated at a resonance. A flash of conscious emotion is not the soma-field; it is the field's Green's function evaluated at a threshold-crossing resonance. The manifolds differ — one is the four-dimensional spacetime vacuum, the other is the eleven-dimensional emotional coupling geometry. The mathematical type is the same. This is not analogy.

2.6 2.6 The Body Schema, Interoception, and Pain

A complete model of the emotional field must address a phenomenon that standard psychological accounts of emotion consistently underspecify: the field is not a model of the physical body. It is the nervous system's *predictive model* of the body — a continuously updated internal representation of what the soma should be experiencing, revised by incoming interoceptive signals.

The clinical proof of this distinction is phantom limb pain [@ramachandran1998]. Patients who have undergone amputation routinely experience pain in the absent limb. The pain is real: it activates the same neural circuits, produces the same suffering, and responds to the same analgesics as pain from an intact limb. The limb is gone. The neural model of the limb persists. What hurts is the *brain's representation* of the foot, not the foot.

This is not an anomaly. It is the normal condition of all somatic experience. The brain does not receive raw signals from the body — it maintains a continuous predictive model of the body (the *body schema*) and generates somatic experience from that model. Interoception — the sense of the internal body state — is a prediction, not a direct readout [@seth2021]. The brain predicts what the heart should be doing, what the gut should feel like, where tension should be. The felt body is the predicted body.

The formal consequence is direct: the soma-field's state vector $\mathbf{e}(t)$ must include **somatic modes** — pain states, regional tension, visceral sensation, proprioceptive activation — alongside emotional modes. These are modes of the same field, governed by the same coupling matrix W. The W_{ij} between fear modes and somatic pain modes is the formal account of why fear amplifies pain, why safety reduces it, and why chronic pain and C-PTSD are highly comorbid. They are not separate conditions sharing a correlation. They are the same attractor architecture operating across emotional and somatic modes simultaneously.

Phantom limb as attractor persistence. An amputated limb's somatic modes do not disappear from W when the limb is removed. The neural model persists. When movement- intention modes are activated — attempting to move the absent foot — foot-sensation modes are co-activated via W. If co-activation exceeds threshold, it is experienced as pain. Ramachandran's mirror box provides visual input that disconfirms the prediction error: new sensory evidence that the limb is moving, reducing coupling-driven co-activation, and therefore reducing the pain. This is $W \to W'$: therapy as structural rewriting of the field.

The load-bearing hyphen. The term *emotional-somatic* in clinical literature is not a stylistic compound. The hyphen marks an ontological claim: emotional states and somatic states are not two separate things that correlate. They are two aspects of the same field. The coupling matrix W is precisely the hyphen, made formal.

Therapeutic implication. Somatic therapies — body scanning, sensorimotor work, EMDR's bilateral stimulation — work not on the physical body but on the brain's model of the body. They provide new interoceptive evidence that updates the prediction. They change W. Therapy does not fix the tissue. It updates the model.

2.7 2.7 Correspondence with Existing Emotion Representations

A reasonable objection to any new framework is: *there is already a great deal of structure out here.* This is true. The emotion research literature contains several well-developed representational systems, and the Soma-Field Model must be positioned relative to them. The short answer is that every existing representation is *descriptive*; the Soma-Field Model is *dynamical.* The longer answer follows.

Categorical taxonomies (Ekman 1972; Plutchik 1980; Parrot 2001) assign names and hierarchical membership to emotional states. They are ontologies in the formal sense: a T-Box of classes and subclass relations. Plutchik's wheel additionally defines a *blend* operation — Love := Joy $\sqcap$ Trust, Awe := Fear $\sqcap$ Surprise — which is precisely the OWL2 `intersectionOf` construction. These systems tell you what to call a state. They do not tell you how a state evolves, or which attractor a system settles in when two mechanisms fire simultaneously.

Dimensional models (Russell 1980; Mehrabian and Russell 1974) embed emotions in a continuous space, canonically Valence × Arousal (the *circumplex*), sometimes extended to Pleasure × Arousal × Dominance. These models capture the *coordinates* of a state. The energy landscape of the Soma-Field Model — the function $H(\mathbf{e})$ over emotion-space — is the dynamical generalisation of the circumplex: the circumplex is a snapshot of positions; the energy landscape is the surface over which the field moves. The stable attractors of H are the emotion categories; their coordinates are the circumplex positions.

Process and appraisal models (Scherer 1999; Frijda 1986; the OCC model of Ortony, Clove and Collins 1988) describe the *sequence of evaluations* through which a stimulus becomes an emotion. They are closer to the Soma-Field dynamics — they include temporal stages — but they are deterministic and single-threaded: one appraisal chain, one output. The Soma-Field replaces this with a parallel field update: all modes evolve simultaneously, governed by the full W matrix.

Music-specific schemas (BRECVEMA, Juslin and Västfjäll 2008; Juslin *et al.* 2011; GEMS, Zentner *et al.* 2008) are the closest antecedents to the present model. The BRECVEMA framework identifies eight distinct psychological mechanisms through which music evokes emotion — Brain stem reflex, Rhythmic entrainment, Evaluative conditioning, Contagion, Visual imagery, Episodic memory, Musical expectancy, Aesthetic judgement — each with distinct evolutionary origins, processing speeds, and neural substrates. These mechanisms are the *object properties* of the emotion-induction ontology: they specify which musical features activate which emotional outputs. Juslin explicitly identifies the open problem: *"Exploring how various musical emotions come about through the interaction of multiple psychological mechanisms is an exciting endeavour that has just begun"* [@juslin2011handbook, p. 638]. The W coupling matrix is the formal answer to that open problem. Where BRECVEMA gives a list of mechanisms with characteristic outputs, the Soma-Field gives the interaction tensor W_{ij} that specifies, with numerical precision, what happens when mechanisms i and j fire concurrently.

The deeper connection is spectral. The *eigenmodes* of W — the directions in emotion-space that evolve independently — are the natural resonances of the soma-field: the patterns the field rings with when struck. BRECVEMA mechanisms are inputs: they excite specific rows of W. The eigenspectrum of W is the response: the set of frequencies the manifold can sustain. Where BRECVEMA is a taxonomy of *stimuli*, the eigenspectrum of W is a taxonomy of *responses*. Juslin's open problem — how mechanisms interact — is the question of how stimulus-space maps onto eigenmode-space through W. Section 3.3 develops this.

Body maps (Nummenmaa *et al.* 2014) map emotions to their somatic distribution — where in the body each emotion is felt. These are precisely the spatial support of the soma-field modes: the field configuration corresponding to an attractor state is the body map of that emotion. Body maps are measurements of the attractors; the Soma-Field is the dynamical system that generates them.

The formal correspondence table extends Table 2 to include these systems:

Existing representation	What it captures	Soma-Field equivalent
Ekman categories	Attractor labels (names)	Values of $\mathbf{e}$ at energy minima
Plutchik dyads ($A \sqcap B$)	Blend attractors	Metastable states between two energy minima
Russell circumplex	Coordinates (valence, arousal)	Projection of $H(\mathbf{e})$ onto two axes

Existing representation	What it captures	Soma-Field equivalent
OCC appraisal tree	Single-path sequential process	Single trajectory in the full field
BRECVEMA mechanisms	Object properties: stimulus $\to$ emotion	Rows of W: mechanism i activates mode j
Body maps (Nummenmaa)	Spatial support of each attractor	Modal structure of $\mathbf{e}$ at each minimum

None of these correspondences require modifying either the existing representations or the Soma-Field Model. They are consequences of the model's structure. The formal machinery for exploring these correspondences — typing BRECVEMA mechanisms as Lean inductive constructors, Plutchik blends as type intersections, mechanism profiles as decidable propositions — is developed in the companion file `src/EmotionOntology.lean`.

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3 3. The Soma-Field Model

The field is primary. The felt emotion is secondary — it is what registers when the field is probed. This is the same ontological relationship as between a quantum field and a particle: the field exists continuously and everywhere; the particle is what you observe at the moment of measurement. The Soma-Field Model does not describe what emotions are *made of*. It describes the manifold whose impulse response *is* conscious emotional experience.

3.1 3.1 Emotions as a Persistent Wave Field

The foundational claim of the Soma-Field Model is simple: emotions are not events. They are a *field* — a distributed, continuous quantity defined over the entire soma (body-mind system) at all times.

This field has two coupled components:

1. **The somatic wave $\mathbf{E}_{\text{body}}(x, t)$**: distributed across the body as patterns of visceral sensation, muscle tone, proprioception, interoception, and autonomic state.
2. **The neural wave $\mathbf{E}_{\text{neural}}(x, t)$**: distributed across the nervous system as patterns of activation in cortical, subcortical, and peripheral neural circuits.

These two components are not separate systems. They are coupled — each continuously influencing the other. The total emotional field is their combined state:

$$\mathbf{E}(x, t) = \mathbf{E}_{\text{body}}(x, t) \otimes \mathbf{E}_{\text{neural}}(x, t)$$

The field is characterised by:

- **Multiplicity**: multiple emotional modes can be simultaneously active and interfering
- **Continuity**: it exists at all times, not only during episodes of conscious feeling
- **Spatial distribution**: different aspects of the field are localised in different regions of the soma (the familiar clinical observation that grief is felt in the chest, fear in the gut, anger in the jaw and fists)
- **Temporal dynamics**: the field evolves continuously, driven by the energy function

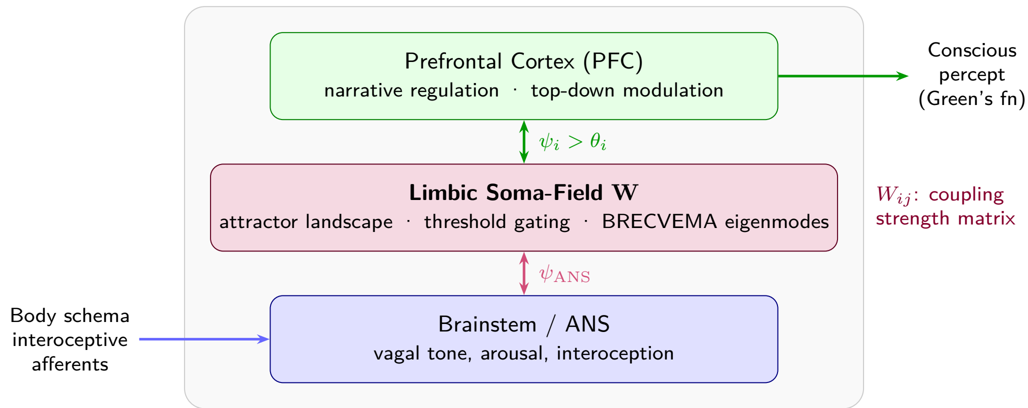

Figure 1. The Soma-Field. The body and brain are not separate containers of emotion but two coupled components of a single distributed wave field. Neither is primary; each continuously modifies the other. The symbols indicate that wave activity is always present in each region, not only during episodes of conscious feeling.

3.2 3.2 The Perception Threshold

Not all activity in the emotional field is consciously perceived. The field has a **perception threshold** T_i for each emotional mode i. Below this threshold, the emotional mode is sub-perceptual: it exists, it influences behaviour and physiology, but it does not surface as a named conscious feeling.

$$\text{Emotion } i \text{ is consciously perceived} \iff |\mathbf{E}_i(t)| > T_i$$

This threshold crossing corresponds precisely to the QFT excitation analogy: the emotional mode behaves like a virtual particle that has accumulated enough energy to become real — to emerge from the sub-threshold background and enter awareness.

This accounts for a range of clinically significant phenomena:

Clinical Observation	Soma-Field Account
Patient reports no feeling but shows physiological signs of distress	Sub-threshold field activity below T_i
Sudden unexpected flood of emotion in session	Rapid threshold crossing after gradual accumulation
Emotion felt somatically but not named	Threshold crossed in $\mathbf{E}_{\text{body}}$, not yet in $\mathbf{E}_{\text{neural}}$
Alexithymia (difficulty identifying feelings)	Elevated T_i — high threshold requiring more energy to cross
Hypervigilance / emotional flooding	Lowered T_i — reduced threshold, field crosses to conscious easily

Table 1. Clinical observations mapped onto the perception threshold model.

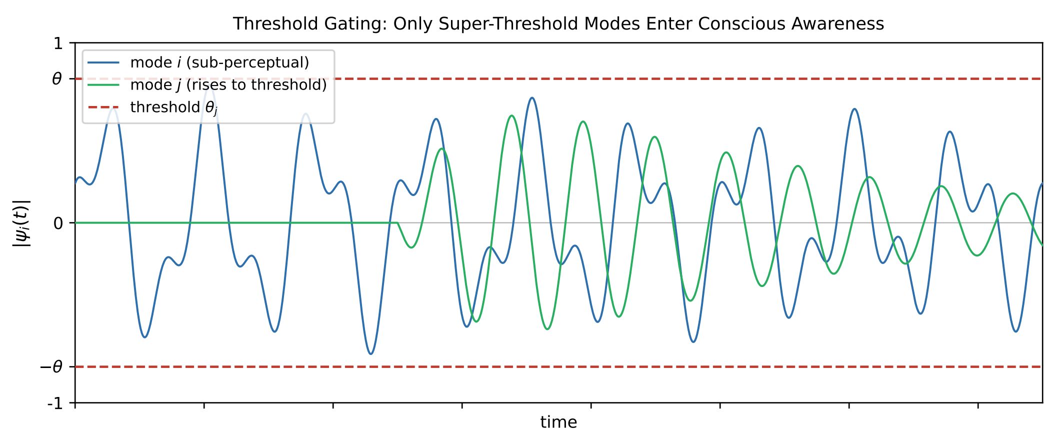

Figure 2. The perception threshold T_i for a single emotional mode. The field is active continuously (lower trace). Conscious experience arises only when amplitude exceeds T_i (upper trace). Everything below the line is still there — shaping body and behaviour before it can be named.

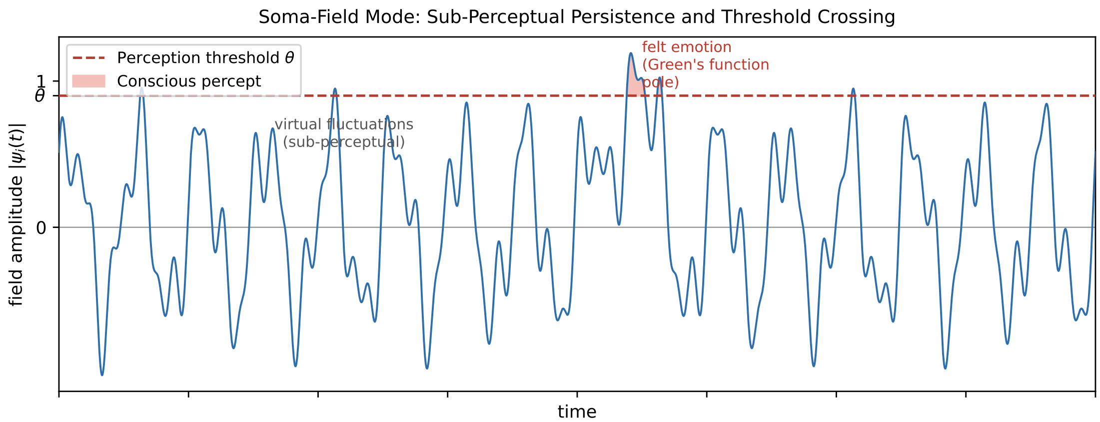

Figure 0. Continuous soma-field activity (blue) with a single threshold-crossing event. The field is always active; conscious experience (shaded) arises only when amplitude exceeds the perception threshold (red dashed). Below the threshold: real, causally active, but not yet conscious.

3.3 3.3 The Interaction of Emotional Modes

Multiple emotional modes are simultaneously active in the field at all times. They do not simply co-exist: they interact. The nature of these interactions is encoded in the **emotional coupling matrix** W, where W_{ij} represents the influence of emotional mode j on emotional mode i.

- If $W_{ij} > 0$: emotion j amplifies emotion i (e.g., fear can amplify shame)
- If $W_{ij} < 0$: emotion j suppresses emotion i (e.g., calm suppresses anxiety)
- If $W_{ij} = 0$: emotions i and j are independent

The field evolves according to the energy gradient:

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}) + \eta(t)$$

where $\eta(t)$ represents the continuous low-level fluctuations of the sub-perceptual field

— the emotional equivalent of quantum vacuum noise. The field is always moving, always seeking lower energy, never at absolute rest.

3.4 3.4 The Three-Layer Architecture

The nervous system that implements the soma-field is not architecturally flat. Three hierarchically organised layers contribute to field dynamics, each corresponding to a distinct evolutionary substrate and a distinct role in the model. The clinical literature (Porges, 2011; van der Kolk, 2014; Ogden et al., 2006) converges on this stratification; what follows is its formal expression.

Layer 1 — Brainstem / autonomic baseline. The oldest structures: vagal nuclei, arousal systems, interoceptive machinery. In the model, this layer is represented by the noise term and, specifically, by the heart rate variability coherence C_{HRV}, which modulates effective noise amplitude across the whole field:

$$\sigma_{\mathrm{eff}} = \frac{\sigma_0}{C_{\mathrm{HRV}}}$$

High HRV coherence narrows effective noise, stabilising the field in its current attractor. This is the mechanism of HRV biofeedback as a regulatory intervention: it does not target any specific emotional mode but lowers the fluctuation floor of the entire field.

Layer 1 extension: cardiac acceleration and landscape tilt. The term C_{HRV} measures the *current state* of cardiac regularity — where the heart is. A complementary quantity is $\dot{H}(t)$, the first time-derivative of heart rate, in units of beats/s^2. This is the **cardiac acceleration**: not what the heart rate is, but where it is going.

The dimensional parallel with gravity is exact: gravitational acceleration g carries units m/s^2; cardiac acceleration $\dot{H}$ carries units beats/s^2. Both are accelerations; both describe a force field rather than a position. Gravity does not tell you where a test mass is — it tells you how it will move next. Cardiac acceleration tells you not the current BPM but the direction of the next one: the N+1 state.

In the soma-field, $\dot{H}(t)$ enters the dynamics not as noise modulation but as a **landscape tilt** — a time-varying bias added to the Hamiltonian that tips the energy function toward activation or rest attractors:

$$H(\mathbf{e}, t) = H_0(\mathbf{e}) - \alpha\, \dot{H}(t)\, \beta \cdot \mathbf{e}$$

where $\alpha > 0$ is the cardiac-somatic coupling constant and β is a mode-coupling vector (at leading order, $\beta = \mathbf{1}$: the tilt acts uniformly across all modes). When $\dot{H}(t) > 0$ (heart accelerating), the landscape tilts toward higher activation states before any cognitive or affective threshold is crossed. When $\dot{H}(t) < 0$ (heart decelerating), it tilts toward rest. The full three-layer equation including the cardiac acceleration term is:

$$\dot{\mathbf{e}}(t) = -\nabla H_0(\mathbf{e}) + \alpha\, \dot{H}(t)\, \beta + R_{\text{PFC}}(\mathbf{e}, t) + \frac{\sigma_0}{C_{\text{HRV}}}\, \xi(t)$$

The two cardiac terms serve distinct functions: C_{HRV} (state) modulates the noise floor; $\dot{H}$ (acceleration) tilts the deterministic landscape. Both are needed for a complete account of cardiac influence on the field.

Predictive clinical value. A patient with BPM = 90 and $\dot{H} = +4$ beats/s^2 is approaching threshold; one with BPM = 90 and $\dot{H} = -4$ beats/s^2 is retreating from it. The snapshot is identical; the trajectories are opposite. Cardiac acceleration is therefore an early-warning signal for threshold crossings — detectable at Layer 1 before the emotional field at Layer 2 has crossed its threshold. This has independent support in cardiology: Bauer et al. (2006) demonstrated that *acceleration capacity* and *deceleration* *capacity* of heart rate — estimates of $\dot{H}$ over a cardiac window — carry prognostic information independent of conventional HRV measures.

The somatic equivalence principle. The cardiac acceleration term $\alpha\,\dot{H}\,\beta$ is structurally identical in the equation to any other forcing term. From the perspective of the field itself — from conscious experience — cardiac-driven activation is indistinguishable from event-driven activation. A sudden heart rate acceleration tilts the landscape by exactly the same mechanism as an external threat or an intrusive memory. The field has no access to the origin of the tilt. This is the formal account of a clinically well-documented phenomenon: anxiety initiated by cardiac irregularity (arrhythmia, postural hypotension, caffeine, exertion) is experienced as emotionally caused, because the somatic signal is identical. Disambiguation requires either external measurement or deliberate interoceptive inquiry that can distinguish the two sources.

Layer 2 — Limbic system / emotional memory. The primary substrate of the Soma-Field Model. The coupling matrix W, memory kernel $K(\tau)$, Hamiltonian $H(\mathbf{e})$, and threshold T all belong here. The limbic layer stores emotional-somatic states and reinstates them in response to partial body cues: a continuous, asymmetric, temporally extended Hopfield network operating on somatic states rather than cognitive patterns. This is the architectural layer that has been absent from every artificial neural network since McCulloch and Pitts (1943) [@mcculloch1943]. The cortex has been modelled many times; the limbic system has not.

Structural plasticity under adversity. The Soma-Field framework permits a formal characterisation of the field's resilience under adverse conditions. Define the *plasticity index* Π as a composite of three measurable field properties:

$$\Pi \;=\; \frac{1}{S_{\text{inst}}} + \left.\frac{\partial \|W\|}{\partial t}\right|_{\text{adversity}} + C_{\text{HRV}}^{\text{recovery}}$$

The three terms correspond to: (i) how accessible regulated-state attractors remain under adversity ($1/S_{\text{inst}}$, instanton accessibility — Section 4.4); (ii) how much the coupling matrix can structurally adapt following a threshold crossing ($\partial\|W\|/\partial t$, the plasticity component); and (iii) how quickly the HRV floor recovers after activation ($C_{\text{HRV}}^{\text{recovery}}$, the regulatory resilience component). Complex PTSD is the clinical presentation of chronically low Π across all three terms simultaneously: high barriers to regulated attractors, a rigid W dominated by threat configurations, and impaired C_{HRV} recovery. Structural plasticity is the capacity of the field to update W in the aftermath of adversity without the adversity permanently *becoming* W.

Layer 3 — Neocortex / prefrontal regulatory layer. Top-down modulation of Layer 2, represented as a regulatory term $R_{\text{PFC}}(\mathbf{e}, t)$. The full field dynamics becomes:

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}(t)) + R_{\text{PFC}}(\mathbf{e}, t) + \frac{\sigma_0}{C_{\text{HRV}}}\,\xi(t)$$

R_{PFC} represents voluntary attention, therapeutic technique, and conscious reappraisal acting on the field. It is not a correction of Layer 2 but a modulation of it. Under sustained therapeutic engagement, R_{PFC} participates in the structural modification $W \to W'$ constituting the forward transformation (Section 7).

The **threshold T is the Layer 2 / Layer 3 boundary**: sub-threshold dynamics are processed limbically and remain below conscious awareness; threshold-crossing events enter Layer 3 and become available for narrative, meaning-making, and voluntary response. This is the formal basis for the clinical observation that insight without somatic activation is limited, and somatic activation without Layer 3 engagement cannot produce structural change: the layers are coupled, not independent. R_{PFC} requires a threshold crossing in order to have something to work with.

The two-term Langevin equation introduced in Section 3.3 is the Layer 2 special case

($R_{\text{PFC}} = 0$, $C_{\text{HRV}} = 1$). All subsequent sections develop that special case. The full three-layer equation is the general form.

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4 4. The Energy Landscape

4.1 4.1 The Structure of the Emotional Energy Function

The energy function $H(\mathbf{e})$ defines a landscape over the space of possible emotional states. Like a physical landscape of hills and valleys, this landscape has:

- **Valleys (local minima)**: stable emotional states the field naturally moves toward
- **Hills (local maxima)**: unstable configurations the field naturally moves away from
- **Saddle points**: transitional configurations with mixed stability

The key property of an energy function is directionality: the field always moves *downhill*. It always evolves toward lower energy. Therapeutic intervention, in this framework, can be understood as:

1. **Changing the landscape**: modifying W — the coupling matrix — through new relational experience, insight, or somatic work, so that the energy minima are in healthier locations
2. **Adding energy to escape a trap**: helping the field accumulate enough energy to escape a deep but unhealthy local minimum (e.g., the freeze state)
3. **Pointing toward the global minimum**: orienting the field toward regulated calm

4.2 4.2 Attractor States: Fight, Flight, Freeze, and Regulated Calm

The Soma-Field Model proposes that the major attractor basins of the emotional energy landscape correspond directly to the autonomic states described by Porges' polyvagal theory.

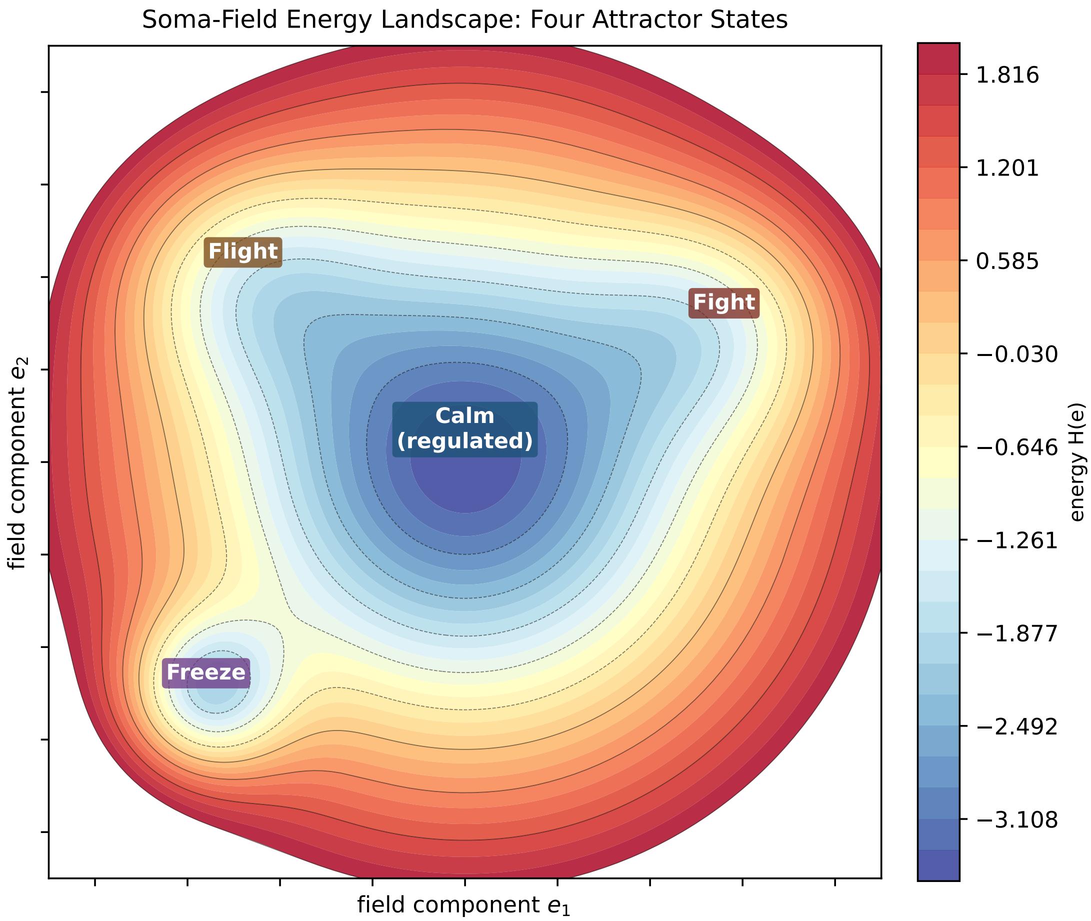

Figure 3a. Topographic (bird's-eye) view of the energy landscape. The field always rolls downhill toward the nearest minimum. Freeze and calm are both low-energy — but freeze is surrounded by high walls. Escape from freeze requires crossing those walls, which means first gaining energy before losing it again. This is the clinical challenge of working with dissociative states.

```
ENERGY
   │
 H │      fight/flight
   │      ┌──┐  ┌──┐
   │      │  │  │  │
   │  ____│  │  │  │____
   │  │      \/       │
   │  │     saddle     │
   │  │  (transition)  │
   │  │                │   ╔══════════╗
   │  │     freeze     │   ║          ║
   │  │      ┌──┐      │   ║ regulated║◄── global minimum
   │  │______│  │______│   ║   calm   ║
   │            │          ╚══════════╝
   └──────────────────────► EMOTIONAL STATE SPACE
```

Figure 3b. Schematic energy landscape. Fight/flight are high-energy, unstable local minima. Freeze is a low-energy but isolated attractor — easy to enter, hard to escape. Regulated calm is the global energy minimum.

Attractor	Energy State	Polyvagal Correlate	Clinical Presentation
Regulated Calm	Global minimum	Ventral vagal (social engagement)	Present, flexible, connected
Fight	Shallow high-energy minimum	Sympathetic (mobilisation)	Agitation, anger, urgency

Attractor	Energy State	Polyvagal Correlate	Clinical Presentation
Flight	Saddle point / shallow minimum	Sympathetic (mobilisation)	Anxiety, avoidance, rumination
Freeze	Deep isolated minimum	Dorsal vagal (immobilisation)	Dissociation, numbness, collapse

Table 2. Emotional attractors mapped onto Polyvagal states.

The therapeutic significance of this structure is considerable. The freeze state is dangerous not because it is high-energy — it is in fact very low energy — but because it is *isolated*: surrounded by energy barriers that make it difficult to exit. Escape from freeze requires first *increasing* the field's energy (mobilising some arousal) before it can flow toward regulated calm. This corresponds well to the clinical observation that working with dissociated patients requires careful titration of arousal — not too much, not too little — before emotional processing is possible.

4.3 4.3 The Coupling Matrix as a Personal Signature

The coupling matrix W is not universal. Each person has a unique W, shaped by attachment history, trauma, cultural context, and temperament. A person with a history of developmental trauma may have a W in which anxiety and shame are strongly coupled ($W_{\text{shame, anxiety}} \gg 0$), creating a combined attractor that is particularly deep and sticky. A person with a secure attachment history may have a W in which positive emotions are broadly coupled to one another, creating a wide basin around regulated calm.

This implies that the energy landscape is a therapeutic object in its own right: understanding a patient's W is understanding the structural dynamics of their emotional life.

In the M-theory compactification analogy developed in Appendix A, the coupling topology W corresponds to the shape of the compact G_2 manifold — the seven-dimensional geometry that determines which force-like couplings are allowed and with what strengths. That analogy is here made precise: two people differ not merely in their emotional *parameter settings* but in their coupling *geometry*. Developmental trauma does not set a dial to the wrong value; it deforms the manifold. The therapeutic process of modifying W through relational experience, insight, or somatic work is, in this language, differential geometry: a continuous deformation of the G_2 manifold toward a configuration in which the regulated-calm attractor is globally accessible. The practitioner is, without having been told so, a geometer.

---

5 5. Dissonance and Resolution

5.1 5.1 The Acoustic Analogy

The Soma-Field Model draws a further structural analogy, this time with acoustics. When two sound waves interact, the quality of their interaction — consonance or dissonance — depends on the phase relationship between them. Consonant intervals (the octave, the fifth) have simple frequency ratios and produce stable, reinforcing interference patterns. Dissonant intervals (the tritone, the minor second) have complex ratios and produce beating, unstable, tension-generating patterns.

The model proposes that the same relationship holds between emotional modes. When two emotional modes are in a compatible relationship — when their interaction is consonant — the field is in a relatively low-energy configuration and moves naturally toward the energy minimum. When they are in an incompatible relationship — when their interaction is dissonant — the field is in a higher-energy configuration, generating a gradient that drives toward resolution.

Dissonance, in this framework, is felt as tension. It is not pathological; it is directional. Dissonance is the field's way of communicating that it is far from equilibrium and that resolution is available.

5.2 5.2 The Resolution Principle

In music, dissonance resolves to consonance. The tritone — the most dissonant interval in Western tonality — creates a powerful gravitational pull toward resolution. In counterpoint, the rules of voice leading describe the specific paths by which dissonance must resolve. These rules are not arbitrary conventions; they describe the geometry of the acoustic energy landscape.

The same principle applies to emotional dissonance. An unresolved emotional state — grief that has not been fully experienced, anger that has been suppressed, fear that has been dissociated — is a dissonance in the field. It generates a persistent tension gradient. The therapeutic process can be understood as guided voice leading: finding the specific path of resolution that transforms the dissonant configuration into a consonant one.

This provides a formal basis for a widely-held clinical intuition: that emotions need to be *felt through* rather than avoided. Avoidance keeps the field in a dissonant state. The energy minimum — regulated calm — lies on the other side of the dissonance, not around it.

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6 6. The Soma-Field Instrument

6.1 6.1 Rationale

The Soma-Field Model is not only a theoretical framework. It motivates a practical therapeutic instrument: a means by which a person can *externalise* their emotional field — make it visible and audible — and interact with it in real time.

The core insight is that the emotional field is normally invisible to its host. It operates below the threshold of conscious awareness, shaping behaviour and physiology without being available for reflection. If its activity could be rendered as a signal — a sound, an image, a pattern — it could become an object of therapeutic attention.

6.2 6.2 Design

The instrument uses a MIDI controller with 16 rotary knobs as its input interface. Eight emotional dimensions are encoded, each represented by two knobs:

- **Knob 1** of each pair: the somatic (body-level) intensity of that emotional mode
- **Knob 2** of each pair: the cognitive/neural intensity of that emotional mode

This design reflects the two-component structure of the field: body and mind are encoded separately but coupled in the computation. Each knob has a continuous range, allowing fine expression of emotional intensity.

```
┌─────────────────────────────────────┐
│         MIDI CONTROLLER             │
│                                     │
│  [K1][K2]  [K3][K4]  [K5][K6]  [K7][K8]  │
│  emotion1  emotion2  emotion3  emotion4│
│                                     │
```

```
│  [K9][K10] [K11][K12][K13][K14][K15][K16] │
│  emotion5  emotion6  emotion7  emotion8│
└──────────────────────────────────────┘
                  │
                  ▼
        ┌───────────────────┐
        │  ENERGY FUNCTION  │
        │  H(e) computed    │
        │  ∇H(e) computed   │
        └───────────────────┘
                  │
     ┌────────────┼────────────┐
     ▼            ▼            ▼
  AUDIO OUTPUT    MIDI OUTPUT     VISUAL OUTPUT
(timbre reflects  (pitch/velocity  (field map:
 dissonance)       reflects energy)  wave topology)
```

Figure 4. The Soma-Field Instrument: input, computation, and multimodal output.

6.3 6.3 The Feedback Loop

The instrument creates a **closed feedback loop** between the person and their emotional field:

1. The person expresses their current emotional state by adjusting the knobs
2. The system computes the energy function $H(\mathbf{e})$ and its gradient ∇H
3. The energy level, dissonance, and proximity to attractor states are rendered as:
   - **Sound**: harmonic content and timbre reflect the consonance or dissonance of the current state

- **MIDI output**: pitch rises with tension, resolves as energy decreases
- **Visual**: a real-time map of the emotional field, showing wave activity, threshold crossings, and the direction of the energy gradient

4. The person hears and sees their emotional field, and adjusts the knobs in response

This loop externalises the emotional field's gradient — the direction in which it is *trying* to move — and makes it available as sensory information. The person becomes not only the source of the emotional signal but also its observer, creating the conditions for reflection and regulation that are at the heart of therapeutic work.

6.4 6.4 The Pluggable Emotion Model

No single model of the emotions is assumed. The coupling matrix W — the structure that determines how emotional modes interact — is loaded from an external configuration file. Standard models (Plutchik's wheel of emotions, Ekman's basic emotions, the valence-arousal-dominance dimensional model) are provided as defaults. The therapist or client can modify the coupling values to reflect their own understanding of their emotional patterns, or a new model can be substituted entirely. The computational engine is model-agnostic.

---

7 7. Clinical Implications

7.1 7.1 Assessment

The Soma-Field Model suggests a different orientation for emotional assessment. Rather than asking "What emotion do you feel?" — which presupposes threshold-level conscious awareness — it invites attention to the sub-perceptual field: "What is present in the body right now, even if you cannot name it?" This aligns with

Focussing-oriented approaches and with sensorimotor methods that prioritise somatic signal over narrative content.

The energy landscape provides a clinical map. A person chronically in a fight or flight attractor shows a different energy signature from a person in a freeze attractor, even if their presenting narratives are superficially similar. The model suggests that these are structurally different therapeutic challenges: fight/flight require down-regulation, while freeze may first require careful up-regulation before down-regulation becomes possible.

7.2 7.2 Intervention

The energy function provides a formal basis for several existing clinical interventions:

- **Grounding and titration** (Levine, 2010): adding small, controlled amounts of energy to the field to approach — without flooding — a previously frozen or avoided emotional state
- **Pendulation** (Levine, 2010): oscillating between a dissonant state and a resource state, progressively widening the tolerance window — equivalent to approaching the energy minimum via a series of small excursions
- **Somatic resourcing** (Ogden et al., 2006): establishing a stable low-energy region in the landscape that the field can return to after excursions into high-energy territory
- **Working with the felt sense** (Gendlin, 1978): attending to sub-threshold field activity and allowing it to cross the perception threshold in a supported context

7.3 7.3 Psychoeducation

The wave model is immediately accessible to clients who have struggled to understand their emotional experience. The statement: *"Your emotions are like waves — they* *are always there, even when you can't feel them, and they are always moving"* is both technically accurate within the Soma-Field framework and clinically useful as a normalising frame for sub-threshold emotional activity, for the apparently sudden onset of strong feelings, and for the experience of feeling multiple conflicting emotions simultaneously.

The energy landscape metaphor — *"right now the field is in a valley that is hard to leave, but it is not the lowest valley available to you"* — offers a way to discuss the freeze state, dissociation, and emotional stuckness without pathologising, while still acknowledging the structural difficulty of these states and the work required to shift them.

7.4 7.4 Neurodivergent Conditions as Operator Modifications

A clinically significant extension of the Soma-Field Model concerns neurodivergent conditions — specifically Autism Spectrum Condition (ASC), Attention Deficit Hyperactivity Disorder (ADHD), and Complex Post-Traumatic Stress Disorder (C-PTSD), which frequently co-occur and each present distinct challenges for somatic emotional processing.

The key architectural principle is this: **these conditions are not parameter settings in the model. They are structural modifications to the operators themselves.** This distinction matters both mathematically and clinically. A parameter change ("set the fear threshold lower") is a quantitative adjustment within the existing structure. An operator modification changes the *form* of the dynamics — it alters the governing equations, not merely their coefficients. Each condition wraps the standard pipeline in a different functional modifier, and — critically for the many individuals who carry all three — these modifiers *compose.* The combined condition is not three separate problems; it is the composition of three operators acting on the same underlying field.

The mathematical details of each modifier are given in Appendix B. Clinically, the consequences are as follows.

Complex PTSD introduces a *memory kernel* into the field dynamics: past high-energy states leave decaying echoes that continue to excite the field without new external stimulus. This is why traumatic activation can appear without identifiable trigger — the field is responding to its own history, not its current environment. The standard Hopfield attractor topology is also disrupted: C-PTSD renders the freeze attractor pathologically deep and wide, the window of tolerance (the basin around regulated calm) pathologically narrow, and the coupling matrix asymmetric — a condition under which the field can enter persistent *limit cycles* rather than settling to a stable minimum. Re-experiencing, flashbacks, and hypervigilance are, in this framework, limit-cycle oscillations in the traumatised field.

```
REGULATED FIELD  (symmetric W, no memory kernel)
──────────────────────────────────────────────────────

|          ╭──────╮                    ╭────────────╮
|   ╭─╮    /       \         ╭─╮     /              \     ╭
|  /   \  /         \  ╭╮   /   \   /                \   /
T -/────\-/──────────\╯-╰/────\-/─────────────────\─/─ T
|        \           ╰──╯       \                   \─╯
|         ╰──────────────────────────────────────────
└──────────────────────────────────────────────────► t

  ↑ baseline returns to near-zero between episodes
  ↑ each threshold crossing is a discrete, independent event
 ↑ 'regulated calm' is a genuine resting state — the global energy minimum
```

```
C-PTSD MODIFIED FIELD  (asymmetric W, memory kernel K(t-s) present)
────────────────────────────────────────────────────────────────────────────

|╭──────────╮              ╭──────────╮              ╭──────────────────────────
T|          \  ╭─╮  /          \  ╭─╮  /                                    ─ T
||           \/   \/            \/   \/
||   ← even the troughs stay near T or above: baseline is elevated
└──────────────────────────────────────────────────────────────────────────► t
   ↑ memory kernel: each activation feeds energy back into the next
  ↑ field rarely returns to true rest — past states re-enter present dynamics
   ↑ almost entirely above T: activation is the default, not the exception
  ↑ 'regulated calm' requires a non-perturbative transition (the instanton):
     small steps do not reach it; a qualitatively different move is needed
```

Figure 5. The same emotional field mode under two dynamic regimes. Top: regulated dynamics — the field oscillates and returns to a low baseline between episodes; conscious emotion (above T) is episodic and resolves. Bottom: C-PTSD-modified dynamics — the memory kernel elevates the baseline so that the field rarely returns to rest; episodes bleed into one another; the system cycles rather than settles. The mathematical basis for this comparison is given in Appendix B.2.

Developmental timing and what can be recovered. The character of the C-PTSD modification depends critically on *when* it occurred — the developmental age τ_d at which the primary traumatic modification took place.

For **late trauma** (τ_d large — adult or post-verbal): a coupling matrix W_0 formed before the event. The modification is additive: $W = W_0 + \delta W_{\text{trauma}}$. A counterfactual pre-trauma self exists, encoded in explicit narrative memory. Therapeutic processing can target δW specifically, and the goal of recovering proximity to W_0 is formally coherent.

For **early trauma** (τ_d small — pre-verbal, perinatal): the coupling matrix W was *formed under the modification*. There is no W_0. The asymmetric coupling and the memory kernel coefficients are the baseline architecture, not additions to one. A counterfactual pre-trauma self was never encoded — it does not exist as a recoverable state.

This is a formal statement of a clinical fact that somatic therapists recognise but rarely have a mechanistic basis for: early trauma cannot be *processed away* in the sense of recovering a prior self, because no prior self was formed. The therapeutic goal is not subtraction ($W \to W_0$, which is undefined) but **forward transformation**: constructing a W' that supports a wider window of tolerance, different attractor topology, and lower memory-kernel amplitudes. This is a different mathematical operation — and requires a different therapeutic model.

The Soma-Field Instrument can reflect this distinction directly: a user whose primary modification is pre-verbal initialises with a *structural* coupling matrix (the modification *is* the baseline), not a neurotypical matrix with an added modifier. The formal basis for this parameterisation is given in Appendix B.2.1.

ADHD raises the effective *thermal noise* of the field — the amplitude of the sub-perceptual fluctuations — and simultaneously reduces the damping coefficient that slows the field's response to the energy gradient. The result is a field that explores its energy landscape rapidly and unpredictably, is easily displaced from shallow attractor basins by small perturbations (distractibility), but also achieves states of intense concentration (hyperfocus) when the coupling to a high-salience stimulus temporarily deepens a specific attractor basin far beyond its resting depth. ADHD is not a deficit of attention; it is a high-temperature, low-damping emotional field with a stimulus-dependent attractor structure.

Autism Spectrum Condition modifies the *projection kernels* — the functions that determine how the continuous somatic field is sampled to produce the discrete state vector — and the *sparsity* of the coupling matrix. Interoceptive research in autism (Garfinkel et al., 2016) documents significant differences in the processing of internal body signals; in model terms, certain somatic regions are over-represented (heightened sensory sensitivity) and others under-represented (reduced interoceptive clarity, contributing to alexithymia). The coupling matrix in ASC tends toward greater sparsity — fewer strong cross-modal emotional couplings — a pattern consistent with monotropism (Murray, 2018): the field settles deeply into individual attractors but transitions between them require proportionally more energy. Intense interests, emotional consistency within a context, and difficulty with unexpected transitions all follow from this attractor topology.

For the Soma-Field Instrument, the practical implication is significant. Rather than asking a neurodivergent user to configure their experience through knob adjustments, the system can instantiate the appropriate operator modifications as a named profile — *"load C-PTSD modifier"*, *"load ADHD modifier"* — each of which transforms the pipeline at the correct mathematical level. The user then interacts with a field that already reflects their structural reality, rather than one calibrated for a neurotypical baseline.

A further clinical implication deserves explicit statement. The Soma-Field Model locates interoceptive accuracy in the field itself: whether a somatic signal has exceeded its perceptual threshold T_i is a property of the field state, not a property of the clinician's assessment of the patient's credibility. A patient reporting an acute somatic state is reporting a threshold-crossing event. The model provides no mechanism by which external disbelief suppresses that crossing. Modified projection operators — as occur in ASC — produce *different* somatic self-reports; the model gives no reason to assume they produce *less accurate* ones. The clinical literature documents a systematic tendency to interpret unusual interoceptive self-reports from neurodivergent patients as indicative of psychogenic origin rather than genuine somatic signal (Nicolaidis et al., 2015). The Soma-Field Model predicts that this interpretive pattern constitutes a category error: it confuses operator modification with signal absence. The practical consequences — missed diagnoses, deferred treatment, and the iatrogenic reinforcement of existing trauma — are well-documented and, within this framework, mathematically predictable.

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8 8. Limitations and Future Directions

The Soma-Field Model is a theoretical framework and must be evaluated as such. Its current form makes several idealisations that require scrutiny.

The coupling matrix W is treated as a fixed parameter, but emotional coupling is dynamic: it changes with context, relationship, and developmental history. A more complete model would treat W as a slowly-evolving quantity, shaped by the field's own history — a form of synaptic plasticity applied to the emotional domain.

The threshold T_i is treated as a fixed property of each emotional mode, but experimental evidence suggests that thresholds are modulated by attentional focus, arousal level, and interpersonal context. A person in a safe therapeutic relationship will typically have lower thresholds — more material reaches conscious awareness — than the same person in an unsafe context.

The acoustic analogy, while structurally productive, requires empirical grounding. The claim that emotional dissonance and acoustic dissonance share formal properties is a hypothesis, not an established finding. Empirical work comparing physiological measures of emotional tension with acoustic analysis of synchronised vocal or musical output would be a productive direction for testing this claim.

The instrument described in Section 6 is a prototype concept. User studies with clinical populations, and collaboration with practising therapists, will be required to assess its therapeutic utility and to identify appropriate clinical contexts.

Future theoretical work should address the relational field: the observation, familiar in systemic and relational approaches to psychotherapy, that emotional fields are not bounded by individual bodies but are co-generated in the space between people. The coupling matrix W of a relationship may be as clinically significant as the W of an individual.

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9 9. Conclusion

The Soma-Field Model proposes a formally grounded account of emotional dynamics that is consistent with the clinical observations of somatic psychotherapy, polyvagal theory, and Focussing-oriented practice. Its central claims — that emotions are a persistent distributed field, that conscious experience is a threshold crossing, and that emotional dynamics are governed by an energy function that drives the field toward stable attractor states — are not novel as clinical intuitions. What is novel is the formal structure that unifies them, and the instrument that the structure motivates.

The model does not resolve the philosophical question of what emotions fundamentally *are*. It offers instead a working representation: one that is precise enough to be computationally implemented, close enough to existing clinical frameworks to be therapeutically applicable, and open enough to be modified as understanding deepens. It invites the therapist to think of the consulting room as a space in which two emotional fields interact — each shaping the other's energy landscape — and of therapeutic work as the art of attending to that interaction with enough precision and care to guide both fields toward lower energy, toward greater coherence, toward regulated calm.

The wave is always there. Therapy is learning to listen to it.

A note on provenance. The Soma-Field Model was not developed from a position of theoretical neutrality. The author carries, as primary data, a lifetime of direct experience of the dynamics described above. The neurodivergent operator modifications of Appendix B are not theoretical abstractions: the C-PTSD memory kernel of B.2 was installed pre-verbally, at approximately eighteen months of age, during a developmental trauma that predates language acquisition entirely. No narrative trace of the origin event exists — there was no verbal capacity with which to encode one. Only the field echo remains, and a measurable physical asymmetry in the body that received it. The ASD and ADHD operator modifications of Appendix B.4 and B.3, respectively, were the instruments by which the model was subsequently constructed: the monotropic attractor structure of B.4 provided the capacity for sustained engagement with an entirely unfamiliar theoretical domain; the high-temperature field dynamics of B.3 drove rapid traversal across it.

The proximate cause is described in full in the companion patient-facing publication. Briefly: an acute somatic emergency in 2025 — a genuine threshold-crossing event, later confirmed as cerebral hypoxia secondary to Long Covid — was attributed, at clinical presentation, to psychiatric origin. The present paper is, among its other functions, a formal response to that attribution.

The causal chain is as follows. A pre-verbal trauma in approximately 1968 installed the C-PTSD operator modifications described in Appendix B.2. The ASD and ADHD modifications of Appendix B.3 and B.4 shaped the system across the intervening decades. Fifty-seven years later, that system's accurate interoceptive signal was dismissed as psychiatric noise. The paper which formally demonstrates that this dismissal constitutes a category error was produced, as a direct causal consequence, by the same operator stack that it describes. The paper is the fixed point of its own subject matter. The author considers this observation methodologically significant.

9.1 9.1 Publication Claim Registry

To support claim-level review rather than all-or-nothing acceptance, this manuscript registers its highest-impact claims with scope labels and disconfirmation tests.

Claim ID	Claim	Scope	Evidence in this work	Disconfirmation criterion
SF-1	Conscious percept is a propagator pole of the soma-field	S1 Structural	Formal derivation in Sections 2-3	Inability to express percept dynamics as Green's-function response under stated operator

Claim ID	Claim	Scope	Evidence in this work	Disconfirmation criterion
SF-2	Emotional attractors are Hopfield-energy minima	S2 Predictive	Energy model and trajectory framework	Constructed update rule under model assumptions with systematic energy ascent
SF-3	Threshold governs felt vs sub-felt emotional activity	S2 Predictive	Threshold operator and clinical mapping	Reliable high-amplitude mode activity with no threshold-dependent behavioural or physiological signature

Claim ID	Claim	Scope	Evidence in this work	Disconfirmation criterion
SF-4	Topological barriers explain classical therapeutic plateaus	S2 Predictive	Formal treatment plus linked companion experiments	Controlled demonstration that matched low-noise classical dynamics crosses registered barriers at equivalent rate
SF-5	Quantum extension yields topological reachability advantage	S2 Predictive	QUANT-EXP-1 companion evidence and linked artifacts	Controlled replication showing no reachability advantage over matched classical baseline

Scope labels: S1 = structural; S2 = predictive; S3 = independently replicated. Current publication target for core claims is S2.

9.2 9.2 Claim-Evidence-Result Matrix

To make review traceable, each core claim is paired with concrete evidence outputs and current result status.

Claim ID	Evidence artifact(s)	Current result status
SF-1	Sections 2-3 derivation of field/propagator structure	structural derivation complete
SF-2	Energy formulation + instrument runtime equations	predictive structure complete
SF-3	Threshold operator definition + clinical interpretation sections	predictive mapping complete
SF-4	Barrier analysis and companion sweep outputs (`paper/QUANT-EXP-SWEEP-2026-05-20.md`)	supported in companion results
SF-5	QUANT-EXP outputs (`instrument/quantum_experiment.py`, `instrument/quantum_sweep_results.csv`)	pilot support: classical cold stuck (Fear approx 0.976, Awe approx 0.000), quantum Awe-dominant occupancy peak approx 0.408

This matrix is intended for reviewer navigation and is updated as companion results are expanded or independently replicated.

9.3 9.3 Replication Package Requirements

To make SF-2 through SF-5 externally testable, each release tagged for review must ship a minimal replication package that can be executed without private context.

Required contents:

1. model code used for the run (`instrument/quantum_experiment.py` and support modules),
2. full parameter snapshot (W, $\mathbf{b}$, γ, D, θ, temperature policy),
3. raw trajectory logs with timestamped attractor labels,
4. analysis scripts that produce the reported summary tables,
5. frozen output artifacts (CSV/plots) referenced in this manuscript.

A claim remains `S2` until an independent operator reproduces directionally consistent outcomes from this package under the same declared protocol.

9.4 9.4 Reviewer-Risk Objections and Responses

To reduce ambiguity in peer review, the highest-probability objections are mapped to bounded responses and concrete upgrade paths.

Reviewer objection	Current response in this manuscript	Remaining action to reach stronger status
"This is an analogy, not a formal model."	Sections 2-4 define operators, dynamics, and testable predictions; Section 9.1 registers disconfirmation criteria claim-wise.	Promote more claims from `S2` to `S3` via independent replication.

Reviewer objection	Current response in this manuscript	Remaining action to reach stronger status
“Evidence is pilot-stage and may not generalize.”	Section 9.2 explicitly labels pilot support and companion-only scope.	Add multi-operator replication and blinded protocol variants.
“Quantum advantage may be implementation-specific.”	SF-5 includes a controlled disconfirmation criterion against matched classical baselines.	Publish full benchmark harness with pre-registered acceptance thresholds.
“Clinical interpretation may exceed data scope.”	Scope labels (`S1`/`S2`/`S3`) and claim registry separate structural from predictive claims.	Add prospective cohort evidence before any clinical-effectiveness claim.

9.5 9.5 Independent Replication Ledger Linkage

`S2` to `S3` promotion for this manuscript is governed by `paper/INDEPENDENT_REPLICATION_LEDGER.md`.

Tracked claim IDs in ledger scope: `SF-2`, `SF-3`, `SF-4`, `SF-5`.

Promotion gate: a claim is upgraded only when at least one ledger row records an independent operator `PASS` with a reproducible package hash and linked raw/derived evidence artifacts.

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11 Appendix A: Categorical Formalization and the M-Theory Scale Hierarchy

This appendix is addressed to readers with a background in mathematics, theoretical physics, or formal methods. It provides the formal structure underlying the clinical model, and describes the type-theoretic foundations required for implementation in a proof assistant such as Lean 4. It can be read independently of the main text.

11.1 A.1 The Missing Layer: Why Category Theory?

The main body of this paper presents the Soma-Field Model in clinical terms. But there is a deeper structural question that the clinical presentation leaves implicit: *how does a model defined at the scale of quantum strings map onto a model of somatic emotional experience?* More practically: *what mathematical language allows each component of* *the system — manifold, field, state, perception, output — to be independently replaced without breaking the pipeline?*

The answer to both questions is the same: **category theory**. A category is a collection of objects and structure-preserving maps (morphisms) between them. A *functor* is a structure-preserving map between categories. The category-theoretic formalization of the Soma-Field Model does three things:

1. Makes explicit the **scale hierarchy** from the 11D emotional manifold to the 1D conscious percept — each step is a functor, and the composition of functors is the full pipeline
2. Defines a **type-theoretically correct interface** at each layer boundary, so that any layer can be replaced (different emotion model, different output renderer, different field representation) without disturbing the others
3. Connects the **holographic principle** to the fractal visual output: both are instances of the same categorical structure — a lower-dimensional boundary encoding a higher-dimensional bulk

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11.2 A.2 M-Theory Compactification as the Scale Model

In M-Theory, the 11-dimensional spacetime is not directly observable. What we experience is a 4-dimensional effective theory, obtained by *compactifying* 7 extra dimensions onto a compact G manifold. Compactification is not a loss of information — it is a reorganisation: the topology of the compactified space determines the particle spectrum and coupling constants of the effective 4D theory. At each scale, a projection functor carries structure downward, integrating out degrees of freedom that are not observable at that level but whose topological invariants are preserved.

The Soma-Field Model proposes an analogous compactification tower for emotional experience:

M-Theory Level	Soma-Field Analogue	Scale
11D M-theory spacetime	Full emotional manifold (all body-mind modes)	Micro (sub-perceptual fluctuations)
7D compact G manifold	Emotional coupling structure (topology of W)	Meso (dispositional patterns)
4D effective spacetime	Observable somatic-emotional field **E**(x,t)	Meso (felt sense, sub-threshold)
1D worldsheet	Conscious percept (single named feeling)	Macro (threshold-crossing)

Table A1. The compactification tower: M-theory levels and their Soma-Field analogues.

G_2 COMPACT MANIFOLD M_7 — schematic (seven compactified dimensions, projected to 2D)

This is the 'hidden' geometry that the full emotional field lives in.

```
      φ¹     φ²     φ³     φ⁴     φ⁵     φ⁶     φ⁷
      |      |      |      |      |      |      |
──────┼──────┼──────┼──────┼──────┼──────┼──────┼──── boundary
      |      |      |      |      |      |      |
┌─────┴──────┴──────┴──────┴──────┴──────┴──────┴──────┐
|     ╭────╮        ╭────╮        ╭────╮        ╭────╮ |
|    /  ·  \       /  ·  \       /  ·  \       /  ·  \|
|   |   ·   |─────|   ·   |─────|   ·   |─────|   ·   ||
|    \  ·  /       \  ·  /       \  ·  /       \  ·  /|
```

```
|     └──┬──┘          └──┬──┘          └──┬──┘          └──┬──┘ |
|        |  \             |  \             |  \             |    |
|        |   └────────────┘   └────────────┘   └────────────┘    |
|                        G₂ holonomy group                         |
|                  (how the dimensions connect)                    |
└──────────────────────────────────────────────────────────────────┘
                                |
        ┌───────────────────────┴──────────────────────┐
        |                                              |
  In M-theory:                                   In Soma-Field:
  topology of M₇                                 topology of M₇
  determines which                               determines the
  particles exist                                coupling matrix W:
  in the 4D theory                               which emotions interact

Each ϕⁱ is one extra dimension, compactified to a tiny circle.
The way these circles connect — the G₂ holonomy — is the structure
of W. Different manifold geometries → different W → different
emotional attractor patterns. Changing the emotion model
(Plutchik → Ekman → custom) is, formally, changing this geometry.
```

Figure A1. Schematic of the compact G manifold M. The seven extra dimensions are represented as circular nodes connected by the G holonomy structure. This diagram is a projection — the actual object is seven-dimensional — but the key point is legible: the topological structure of the connections between dimensions determines the physical properties of the effective theory. In the Soma-Field analogy, it determines W.

The critical structural claim is that the coupling matrix W — the object that determines which emotions amplify or suppress one another — encodes the **topology of the compactified space**. Just as the G holonomy group of the compactified manifold determines which particles can exist in the effective 4D theory, the structure of W determines which emotional configurations are stable in the effective somatic theory. Changing the emotion model (Plutchik → Ekman → custom) is, in this framework, a change of compactification geometry — a different topology, a different set of stable attractors.

11.3 A.3 The Four Categories and Their Functors

Define four categories:

(Manifold Category) - *Objects*: emotional manifold configurations M with G holonomy structure - *Morphisms*: smooth deformations M preserving the G 3-form Ω

(Field Category) - *Objects*: field configurations **E**(x,t) on the soma — sections of a bundle over body × time - *Morphisms*: field evolution operators (propagators, Green's functions)

(State Category) - *Objects*: emotional state vectors **e** with energy function H(**e**) - *Morphisms*: energy-gradient flows — paths **e**(t) satisfying ė = − H(**e**)

(Perception Category) - *Objects*: conscious percepts — threshold-crossing events (emotion i, intensity, timestamp) - *Morphisms*: temporal sequences of percepts (the stream of conscious emotional experience)

The pipeline is a chain of functors:

$$\mathcal{M} \xrightarrow{\Lambda} \mathcal{F} \xrightarrow{\Pi} \mathcal{S} \xrightarrow{M} \mathcal{P} \xrightarrow{O} \mathcal{O}$$

where is any output category (audio, MIDI, visual, haptic — each a separate functor from that can be independently composed).

Λ — Compactification Functor (→) Takes a manifold deformation Ω to a field configuration via the acoustic coupling operator:

$$\Lambda[\delta\Omega] = \frac{1}{2\pi\,\ell_s\,f_0}\int_{M_7} \delta\Omega \wedge \star\,\delta\Omega$$

This is the bridge from M-theory geometry to the observable somatic field. It is also the point at which dissonance (the phase lag Δ between field modes) first becomes computable.

Π — Projection Functor (→) Integrates the continuous field over the soma to produce the finite-dimensional state vector:

$$\Pi[\mathbf{E}](i) = \int_{\text{Soma}} K_i(x)\,\mathbf{E}(x,t)\,dx$$

where $K_i(x)$ is a kernel that weights the somatic region relevant to emotional mode i. This is the step at which the infinite-dimensional field becomes computationally tractable. The choice of kernels K_i is, clinically, the question of *where in the body* each emotion is localised — a question that is configurable and empirically adjustable.

M — Measurement Functor (→) Applies the perception threshold: only states satisfying $|\mathbf{e}_i| > T_i$ are mapped to percepts. Below the threshold, the state exists in but has no image in . This is the QFT excitation step — the collapse from virtual to real.

O — Output Functor (→) Any rendering of conscious percepts as external signal. Each output modality is a separate functor that can be composed, replaced, or run in parallel without touching any upstream component:

```
� ──→ �_audio    (timbre = dissonance, pitch = energy)
� ──→ �_midi     (velocity = |�H|, note = emotional mode)
� ──→ �_visual   (fractal hologram = holographic encoding of field)
� ──→ �_haptic   (vibration pattern = tension gradient)
```

Figure A2. Output functors: each output modality is an independent functor from to its own output category . Adding a new output (e.g., a fractal hologram) requires only implementing a new functor O; no upstream component changes.

11.4 A.4 The Holographic Principle and Fractal Output

The Maldacena AdS/CFT correspondence (1997) establishes that a gravitational theory in an (n+1)-dimensional Anti-de Sitter bulk is dual to a conformal field theory on its n-dimensional boundary. The key claim is that the boundary theory is a *complete* holographic encoding of the bulk: all information about the bulk geometry can, in principle, be reconstructed from the boundary data.

In the Soma-Field context, the holographic correspondence maps as follows:

- **Bulk**: the full emotional field **E**(x,t) — high-dimensional, continuous, distributed
- **Boundary**: the stream of conscious percepts in — lower-dimensional, discrete, local

The holographic principle implies that the boundary data (conscious experience) encodes all the information in the bulk (the full emotional field) — but in a compressed, topologically reorganised form. Much of that information is inaccessible without the right decoding key. Therapeutic work, in this reading, is the development of better decoding — the ability to reconstruct more of the bulk field from the boundary signal.

Fractals as terminal coalgebras. A fractal is the fixed point of a contractive endofunctor $F : \mathbf{Type} \to \mathbf{Type}$. More precisely, it is the *terminal coalgebra* of F — the unique type X such that $F(X) \cong X$. For a fractal visual output rendering the emotional field:

$$F(X) = \text{EmotionalState} \times X$$

The terminal coalgebra of this functor is an infinite stream of emotional states — a coinductive rendering of the field's evolution. Each level of zoom in the fractal corresponds to one level of the compactification tower: the self-similarity of the fractal reflects the self-similar structure of the manifold hierarchy. This is not an aesthetic choice; it is the natural categorical structure of the output.

Any output functor $O : \mathcal{P} \to \mathcal{O}_{\text{visual}}$ that maps to this coinductive type will automatically produce a self-similar rendering. Replacing the fractal with a different visual representation is replacing F with a different endofunctor — the pipeline above it is unchanged.

11.5 A.5 Lean 4 Type Sketches

The following sketches use Lean 4 syntax (Mathlib conventions). They are illustrative rather than complete; `sorry` marks positions requiring additional measure-theoretic or proof-theoretic development.

```
import Mathlib.LinearAlgebra.Matrix.DotProduct

import Mathlib.Analysis.InnerProductSpace.Basic


-- ─────────────────────────────────────────────

-- Core types
```

```
-- ────────────────────────────────────────────

-- Emotional state: n modes, each with body and neural components
structure EmotionalState (n : ⍰) where
  somatic : Fin n → ⍰  -- distributed body signal
  neural  : Fin n → ⍰  -- distributed neural signal

-- Combined activation for mode i
def EmotionalState.activation {n : ⍰} (e : EmotionalState n) (i : Fin n) : ⍰ :=
  e.somatic i + e.neural i

-- ────────────────────────────────────────────
-- The coupling matrix (pluggable emotion model)
-- ────────────────────────────────────────────

structure CouplingMatrix (n : ⍰) where
  W : Matrix (Fin n) (Fin n) ⍰  -- interaction weights
  θ : Fin n → ⍰                 -- activation thresholds

-- ────────────────────────────────────────────
-- Hopfield energy function
-- ────────────────────────────────────────────

def hopfieldEnergy {n : ⍰} (cm : CouplingMatrix n) (e : EmotionalState n) : ⍰ :=
  let s : Fin n → ⍰ := e.activation
  let Ws : Fin n → ⍰ := cm.W.mulVec s
  -0.5 * Finset.univ.sum (fun i => Ws i * s i) -
```

```
  Finset.univ.sum (fun i => cm.θ i * s i)

-- ─────────────────────────────────────────────

-- Perception threshold (dependent type)

-- ─────────────────────────────────────────────

-- A mode is perceived iff its activation exceeds its threshold

def Perceived {n : ℕ} (e : EmotionalState n) (T : Fin n → ℝ) (i : Fin n) : Prop :=

  |e.activation i| > T i

-- The measurement functor: only perceived modes have images

structure Percept (n : ℕ) (e : EmotionalState n) (T : Fin n → ℝ) where

  mode      : Fin n

  evidence  : Perceived e T mode

  intensity : ℝ := e.activation mode

-- ─────────────────────────────────────────────

-- Attractor states (the energy landscape)

-- ─────────────────────────────────────────────

inductive AttractorBasin

  | regulatedCalm  -- global minimum: ventral vagal

  | fight          -- shallow high-energy local minimum

  | flight         -- saddle point

  | freeze         -- deep isolated local minimum: dorsal vagal

-- Basin classification (requires eigenanalysis of W; sketched here)
```

```
noncomputable def classifyAttractor {n : ▯}
    (cm : CouplingMatrix n) (e : EmotionalState n) : AttractorBasin :=
  sorry  -- determined by local curvature of H at e

-- ─────────────────────────────────────────────
-- Output functor interface (pluggable outputs)
-- ─────────────────────────────────────────────

-- Any output module implements this typeclass
class OutputFunctor (α : Type) where
  render : ▯ {n : ▯}, Percept n → α  -- maps a percept to output signal

-- Example instances (implementations live in their own modules):
-- instance : OutputFunctor MidiSignal  -- MIDI output
-- instance : OutputFunctor AudioBuffer -- audio output
-- instance : OutputFunctor FractalFrame -- fractal hologram output

-- ─────────────────────────────────────────────
-- The full pipeline (functor composition)
-- ─────────────────────────────────────────────

-- Project field to state (integration over soma — requires measure theory)
noncomputable def projectField {n : ▯}
    (kernel : Fin n → (Fin 3 → ▯) → ▯)   -- somatic localisation kernels K_i(x)
    (field  : (Fin 3 → ▯) → EmotionalState n) : EmotionalState n :=
  sorry  -- ∫_Soma K_i(x) · field(x) dx
```

```
-- Full pipeline: field → state → percepts → output
def pipeline {n : ⍰} {α : Type} [OutputFunctor α]
    (cm : CouplingMatrix n)
    (T  : Fin n → ⍰)
    (e  : EmotionalState n) : List α :=
  -- collect all perceived modes and render each one
  (Finset.univ.filter (fun i => Perceived e T i)).toList.filterMap
    (fun i =>
      if h : Perceived e T i
      then some (OutputFunctor.render (Percept.mk i h))
      else none)
```

What this buys you. The `OutputFunctor` typeclass is the formal interface for every output module. To add a fractal hologram output, implement `instance : OutputFunctor FractalFrame`. To add haptic output, implement `instance : OutputFunctor HapticPattern`. The pipeline function does not change. The coupling matrix is a struct — to swap emotion models, pass a different `CouplingMatrix`. The `Perceived` predicate is a proposition — Lean's type checker guarantees at compile time that nothing downstream of the measurement functor receives sub-threshold data. These are not software engineering conveniences; they are mathematical theorems enforced by the type system.

11.6 A.6 What Is Not Yet There

Two gaps remain before the formalization is complete:

1. The projection integral. The `projectField` function contains a `sorry` because it requires a full measure-theoretic treatment of integration over a somatic manifold. In practice, for the computational implementation, this integral will be approximated by a weighted sum over discrete sensor readings (MIDI knob values). The Lean proof of this approximation's correctness is non-trivial and is deferred.

2. The compactification functor Λ. The map from G manifold deformations to field configurations (the M-theory-to-field bridge) is stated mathematically in the main text but has no Lean encoding here. A full treatment requires formalising differential forms and G holonomy in Lean — work that is underway in Mathlib but not yet complete enough to build upon directly.

Everything else in the pipeline — the energy function, the perception threshold, the attractor classification, the output routing — is type-theoretically sound and can be compiled today.

11.7 A.7 Reading the Architecture in Terms of This Formalization

Each directory in the project corresponds to a category or a functor:

Directory	Categorical Role
`field/energy.py`	$H : \mathcal{S} \to \mathbb{R}$ — the energy functional
`field/dynamics.py`	Morphisms in $\mathcal{S}$ — gradient flow paths
`field/attractors.py`	Objects in $\mathcal{S}$ that are fixed points of the flow
`models/*.yml`	Alternative coupling matrices — different compactification geometries
`acoustic/coupling.py`	The functor $\Lambda : \mathcal{M} \to \mathcal{F}$

Directory	Categorical Role
`midi/encode.py`	The approximate projection $\Pi : \mathcal{F} \to \mathcal{S}$
`soma/tension.py`	Gradient readout: $\lvert\nabla H\rvert$ as a scalar observable
`visual/field_map.py`	Output functor $O : \mathcal{P} \to \mathcal{O}_{\text{visual}}$
`engine/pipeline.py`	The composition $O \circ M \circ \Pi \circ \Lambda$

Table A2. Project directory structure mapped to categorical roles.

Any new output module — fractal hologram renderer, biofeedback display, inter-body relational field visualisation — is an additional row in the last block of this table: a new functor from $\mathcal{P}$ to a new output category, requiring no changes above it in the tower.

---

12 Appendix B: Neurodivergent Operator Modifications

This appendix provides the mathematical formulation of the three neurodivergent operator modifications introduced in Section 7.4: Complex PTSD, ADHD, and Autism Spectrum Condition. Each is defined as a structural transformation of the standard Soma-Field dynamics. The composition of multiple modifiers is addressed at the end.

12.1 B.1 The Standard Dynamics (Baseline)

The standard Langevin dynamics of the emotional field are:

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}(t)) + \eta(t)$$

where H is the Hopfield energy function, and $\eta(t)$ is white Gaussian noise with amplitude σ_0:

$$\eta(t) \sim \mathcal{N}(0,\, \sigma_0^2 \mathbf{I})$$

The coupling matrix W is assumed **symmetric** ($W = W^\top$), which guarantees that the dynamics have only point attractors — the field always settles to a fixed minimum of H. This symmetry condition is the Hopfield convergence theorem. Neurodivergent modifications break this assumption in specific, characterisable ways.

12.2 B.2 Complex PTSD: The Memory Kernel and Asymmetric Coupling

What C-PTSD adds to the dynamics.

C-PTSD introduces two modifications:

1. A memory kernel (non-Markovian dynamics). The field's evolution is no longer determined solely by its current state; it is influenced by its own history. Past high-energy activations leave exponentially decaying echoes:

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}(t)) + \int_0^t K_{\text{trauma}}(t-s)\, \mathbf{e}(s)\, ds + \eta(t)$$

The trauma memory kernel is a sum of decaying traces, one per encoded traumatic event:

$$K_{\text{trauma}}(\tau) = \sum_k A_k \, e^{-\tau/\tau_k}$$

where $A_k > 0$ is the amplitude (intensity) of the k-th memory trace and τ_k is its decay time (the timescale over which the event's echo fades). **Therapeutic processing reduces A_k and/or increases τ_k** — the echo becomes quieter and shorter-lived. Full processing corresponds to $A_k \to 0$.

2. W asymmetry (enables limit cycles). In C-PTSD, the standard symmetry $W = W^\top$ is broken. The coupling matrix acquires an antisymmetric component:

$$W_{\text{PTSD}} = W_{\text{sym}} + W_{\text{anti}}, \quad W_{\text{anti}} = -W_{\text{anti}}^\top$$

An asymmetric W breaks the Hopfield convergence guarantee: the field may enter **limit cycles** — persistent oscillations between states that never settle to a minimum. Re-experiencing episodes, intrusive affect, and the oscillation between hyperarousal and hypoarousal (the PTSD symptom cycle) are, in this framework, limit cycles generated by the antisymmetric component of W_{PTSD}.

Window of tolerance. The region in state space where regulated calm is stably accessible becomes narrow. Formally: the basin of attraction of the regulated-calm attractor shrinks, while the freeze and fight/flight basins widen. Small perturbations are sufficient to displace the field into a dysregulated state.

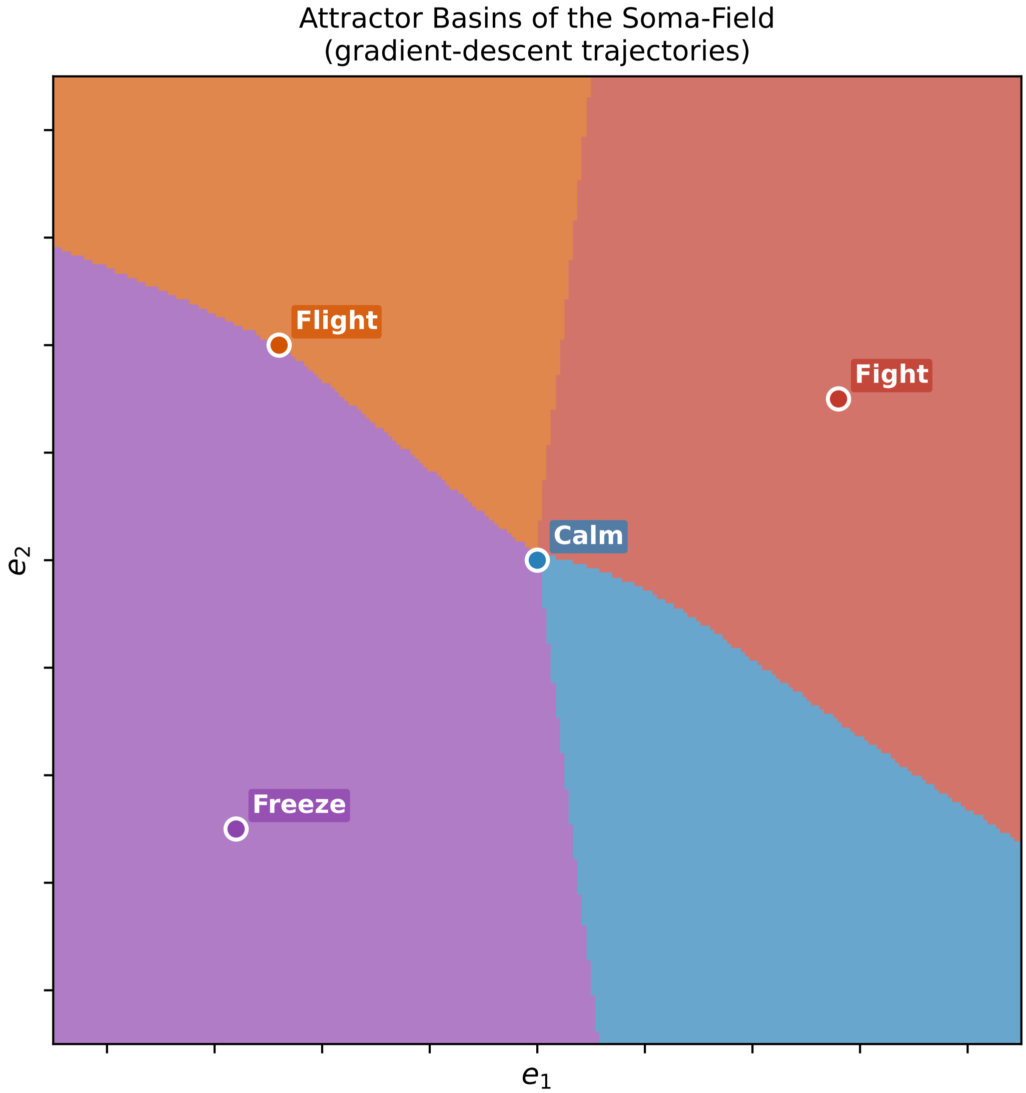

Figure B1. Schematic comparison of the attractor basin topology in standard dynamics (left) and C-PTSD-modified dynamics (right). The calm basin narrows; a limit-cycle orbit appears.

12.3 B.2.1 Developmental Time Parameterisation

Let $\tau_d \in [0, \infty)$ denote the developmental age in months at which the primary traumatic modification occurred. The character of the modification interpolates continuously with τ_d:

$$W(\tau_d) = f(\tau_d) \cdot W_0 + (1 - f(\tau_d)) \cdot W_{\text{trauma}}$$

where W_0 is the neurotypical coupling baseline, W_{trauma} is the asymmetric modification matrix, and f is a smooth interpolation:

$$f(\tau_d) = \tanh\left(\frac{\tau_d}{\tau_c}\right), \qquad \tau_c \approx 36 \text{ months}$$

The critical age τ_c is the approximate onset of verbal encoding capacity — the developmental threshold below which episodic memory is not yet available and traumatic encoding is entirely somatic and procedural.

At $\tau_d = 0$: $f = 0$, $W = W_{\text{trauma}}$ — the modification is fully structural; no neurotypical baseline exists.

At $\tau_d \to \infty$: $f \to 1$, $W \approx W_0$ — the baseline dominates and the trauma is a perturbation on a pre-formed structure.

The memory kernel also depends on τ_d. For pre-verbal trauma ($\tau_d < \tau_c$): traces are somatic and procedural — stored in the body architecture, not in narrative memory. Decay times τ_k tend to be longer (the trace has no verbal reinforcement working against it, but also no verbal processing available to shorten it). For post-verbal trauma: traces have dual encoding — somatic and narrative — and the narrative component is partially accessible through linguistic therapy, though the somatic component persists independently.

In Lean 4:

```
-- Developmental time parameter for the C-PTSD operator
structure TraumaProfile (n : ⍰) where
  τ_d        : ⍰                          -- developmental age at trauma (months)
  asymmetry  : Matrix (Fin n) (Fin n) ⍰  -- antisymmetric W component
  amplitudes : List (Fin n → ⍰)          -- A⍰ per memory trace
  decayTimes : List (Fin n → ⍰)          -- τ⍰ per memory trace

def τ_c : ⍰ := 36  -- verbal encoding threshold (months)

-- Structural fraction: 0 = trauma IS the structure; 1 = trauma added to baseline
noncomputable def structuralFraction (τ_d : ⍰) : ⍰ :=
  Real.tanh (τ_d / τ_c)

-- Coupling matrix interpolated by developmental time
noncomputable def couplingAtDevelopmentalAge {n : ⍰}
    (baseline : CouplingMatrix n) (profile : TraumaProfile n) : CouplingMatrix n :=
  let f := structuralFraction profile.τ_d
  { W := f • baseline.W + (1 - f) • profile.asymmetry,
    θ := baseline.θ }

-- For pre-verbal trauma: no neurotypical W₀ can be recovered by subtraction
-- (the structural fraction is < tanh(1) ≈ 0.76; trauma dominates the coupling)
theorem preVerbalIsStructural {n : ⍰} (profile : TraumaProfile n)
    (h : profile.τ_d < τ_c) :
```

```
    structuralFraction profile.τ_d < Real.tanh 1 := by
  unfold structuralFraction τ_c at *
  exact Real.tanh_lt_tanh_of_lt (by linarith) (by linarith)


-- Corollary: the therapeutic operation for pre-verbal trauma is forward transformation,
-- not recovery.  W → W' with wider window of tolerance;  NOT  W → W₀ (W₀ undefined).
```

What `preVerbalIsStructural` says. For $\tau_d < \tau_c$, the structural fraction is below $\tanh(1) \approx 0.76$: more than 24% of the coupling matrix is trauma-formed rather than baseline-formed. This grows to 100% as $\tau_d \to 0$. The theorem is a formal statement that the goal of recovering a pre-trauma state is not achievable by any subtraction operation on W — because the object that would be recovered (W_0) was never the dominant component. The forward transformation $W \to W'$ is not a second-best option; it is the only coherent one.

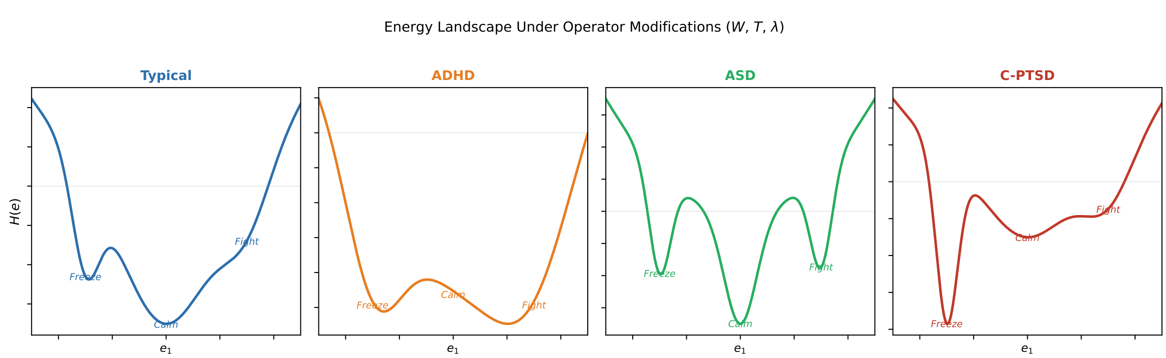

Figure 5. Neurotype comparison: energy landscape modifications for Typical, ADHD, ASC, and C-PTSD dynamics. Each neurotype deforms the attractor topology in a characteristic way.

12.4 B.3 ADHD: High-Temperature, Low-Damping Dynamics

What ADHD adds to the dynamics.

ADHD is modelled as a modification of the Langevin equation's **noise amplitude** and **damping coefficient**:

$$\gamma_{\text{ADHD}}\,\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}(t)) + \sqrt{2D_{\text{ADHD}}}\,\xi(t)$$

with $\gamma_{\text{ADHD}} < \gamma_0$ (reduced damping) and $D_{\text{ADHD}} > D_0$ (increased diffusion / noise temperature).

In statistical mechanics, the ratio $D/\gamma = k_B T_{\text{eff}}$ defines an **effective temperature**. ADHD corresponds to a higher effective temperature: the emotional field explores its energy landscape more rapidly and more randomly. The consequences are:

- **Shallow basins are unstable**: attractor basins below a depth proportional to T_{eff} cannot hold the field. The field is displaced by thermal fluctuations before it can settle — experienced as distractibility and difficulty sustaining attention.
- **Hyperfocus as a deep, stimulus-induced basin**: when a high-salience stimulus is present, the coupling to that stimulus temporarily deepens a specific attractor basin far beyond its resting depth. The field falls into this deep basin and is held there even against competing perturbations — experienced as hyperfocus or flow absorption.
- **Emotional dysregulation (RSD)**: the field's higher temperature means that rejection-sensitive emotional modes can be excited to threshold-crossing amplitude by perturbations that would be sub-threshold at baseline temperature.

The noise $\xi(t)$ in ADHD is not white but exhibits $1/f$ **spectral structure** — low-frequency components carry disproportionate power, consistent with the slow drift of attentional state observed in ADHD time-series data (Gilden, 2001):

$$S_\xi(f) \propto f^{-\alpha}, \quad \alpha \approx 1$$

This $1/f$ (pink noise) structure means that long-range temporal correlations exist in the noise — the field's fluctuations are not memoryless, but have a slow, drift-like component superimposed on rapid fluctuations.

12.5 B.4 Autism Spectrum Condition: Sparse Coupling and Modified Projection

What ASC adds to the dynamics.

ASC is modelled as two modifications at different levels of the pipeline.

1. Modified projection kernels (at the Π functor). The standard projection:

$$\Pi[\mathbf{E}](i) = \int_{\text{Soma}} K_i(x)\, \mathbf{E}(x,t)\, dx$$

uses kernels $K_i(x)$ that determine which somatic regions contribute to the i-th emotional mode. In ASC, consistent with interoceptive research (Garfinkel et al., 2016), these kernels are modified:

$$K_i^{\text{ASC}}(x) = \beta_i(x)\, K_i(x), \quad \beta_i(x) \geq 0$$

where $\beta_i(x) > 1$ in regions of sensory hypersensitivity and $\beta_i(x) < 1$ in regions of interoceptive under-registration. The result is a projection that over-samples some somatic signals and under-samples others, relative to the neurotypical baseline. This accounts simultaneously for sensory sensitivity and for aspects of alexithymia that arise not from absence of the somatic signal but from its modified projection into the named-emotion state vector.

2. Sparse coupling matrix (monotropism). The standard W matrix has many non-zero off-diagonal entries: emotions cross-activate one another broadly. In ASC, the coupling matrix is sparser:

$$W_{\text{ASC}} = W \odot M_{\text{sparse}}, \quad [M_{\text{sparse}}]_{ij} \in \{0, 1\}$$

where $\odot$ denotes element-wise multiplication and M_{sparse} is a binary mask that zeros out many cross-couplings. The attractor topology that results has:

- **Fewer, deeper individual attractors**: each active attractor basin is deeper because the energy is not distributed across many coupled modes
- **Higher inter-attractor barriers**: transitioning between basins requires more energy because cross-coupling pathways are sparse — there are fewer "stepping-stone" intermediate states
- **Monotropic stability**: within a salient attractor (a special interest, a familiar routine, a trusted relationship), the field is highly stable and difficult to displace

This is consistent with the monotropism account of autism (Murray, 2018): attention and interest flow as a stream that fills one channel deeply rather than spreading shallowly across many.

12.6 B.5 Composition: The Three Operators Together

For an individual carrying all three conditions — ASD + ADHD + C-PTSD — the modifiers compose. The combined dynamics are:

$$\gamma_{\text{ADHD}}\,\dot{\mathbf{e}}(t) = -\nabla H_{\text{ASC}}(\mathbf{e}(t)) + \int_0^t K_{\text{trauma}}(t-s)\,\mathbf{e}(s)\,ds + \sqrt{2D_{\text{ADHD}}}\,\xi_{1/f}(t)$$

where H_{ASC} uses W_{ASC} (sparse coupling and modified projection).

The interaction effects are non-trivial and clinically recognisable:

Interaction	Mathematical origin	Clinical presentation
ADHD noise + C-PTSD limit cycles	High-temperature oscillation	Rapid cycling between hyperarousal and shutdown; difficult to titrate
ADHD noise + ASC deep basins	High temperature fights deep wells	Long wind-up before hyperfocus; rapid exit once perturbed
C-PTSD memory + ASC sparse W	Echoes activate narrow pathways	Trauma triggers are specific and apparently disproportionate; hard for others to anticipate
All three	See above, composed	Wide tolerance window required; regulation is genuinely harder, not a matter of effort

Table B1. Interaction effects of composed neurodivergent modifiers.

The architectural point. In the Soma-Field Instrument, loading a neurodivergent profile does not change the knob layout, the emotion model, or the output routing. It wraps the dynamics engine in the appropriate operator modifier. The user interacts with their actual field topology, not a neurotypical approximation of it. Therapeutic progress — processing a trauma trace, extending the window of tolerance, reducing a persistent limit cycle — is visible as a change in the modifier parameters over time: A_k decreasing, the calm basin widening, σ settling.

The model does not pathologise. ASD's sparse coupling is not a deficit; it is a different attractor topology with genuine structural advantages. ADHD's high noise temperature is not a failure to regulate; it is a field that explores its landscape at high speed, with different costs and affordances from a low-temperature field. C-PTSD modifies the dynamics in ways that were, at the time of the trauma, adaptive; the therapeutic task is not to remove the modification but to reduce its amplitude where the original adaptive function no longer serves.

In Lean 4, each modifier is a structure that composes cleanly:

```
structure NeurodivergentModifier (n : ℕ) where
  -- C-PTSD: memory kernel coefficients per mode (empty list = no trauma modification)
  traumaTraces  : List (Fin n → ℝ × ℝ)   -- (amplitude Aₖ, decay time τₖ) per mode
  wAsymmetry    : Matrix (Fin n) (Fin n) ℝ -- antisymmetric W component; zero = no PTSD
  -- ADHD: noise and damping modifications (1.0 = neurotypical baseline)
  noiseTemp     : ℝ   -- D_ADHD / D_baseline; >1.0 = higher temperature
  dampingCoeff  : ℝ   -- γ_ADHD / γ_baseline; <1.0 = less damping
  -- ASC: projection kernel scaling and coupling sparsity mask
  kernelScale   : Fin n → (Fin 3 → ℝ) → ℝ  -- β_i(x) modifier
  couplingMask  : Matrix (Fin n) (Fin n) Bool -- W sparsity; true = keep, false = zero
```

```
-- Compose two modifiers (order matters for non-commuting terms)
def NeurodivergentModifier.compose {n : ℕ}
    (m1 m2 : NeurodivergentModifier n) : NeurodivergentModifier n where
  traumaTraces := m1.traumaTraces ++ m2.traumaTraces
  wAsymmetry   := m1.wAsymmetry + m2.wAsymmetry
  noiseTemp    := m1.noiseTemp * m2.noiseTemp
  dampingCoeff := m1.dampingCoeff * m2.dampingCoeff
  kernelScale  := fun i x => m1.kernelScale i x * m2.kernelScale i x
  couplingMask := fun i j => m1.couplingMask i j && m2.couplingMask i j

-- Named profiles (load from config; override any field as needed)
def Modifier.cptsd   (n : ℕ) : NeurodivergentModifier n := sorry  -- from clinical params
def Modifier.adhd    (n : ℕ) : NeurodivergentModifier n := sorry
def Modifier.asc     (n : ℕ) : NeurodivergentModifier n := sorry

-- Compose all three: the modifier for ASD + ADHD + C-PTSD
def Modifier.compound (n : ℕ) : NeurodivergentModifier n :=
  (Modifier.asc n).compose ((Modifier.adhd n).compose (Modifier.cptsd n))
```

The `compose` function is the formal statement that the three conditions are not additive in their effects — they compose, and the composition order matters where the operators do not commute (specifically, the asymmetric W and the sparse coupling mask interact non-trivially). This is not a software engineering detail. It is a clinical prediction: the joint presentation of ASD + ADHD + C-PTSD is not the sum of its parts, and the Soma-Field Model gives a precise account of why.

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13 Appendix C: String Diagrams and the Cross-Language Correspondence

This appendix introduces string diagram notation and demonstrates that the Soma-Field Model's core claims are simultaneously expressible — and simultaneously legible — in theoretical physics, category theory, Lean 4 type theory, and somatic clinical description. The notation used here is due to Penrose (1971) and formalised by Selinger (2010, "A Survey of Graphical Languages for Monoidal Categories"); the identification of string diagrams with Feynman diagrams in the context of categorification is due to Baez and Lauda (2011, "A Prehistory of n-Categorical Physics"). Readers unfamiliar with physics or formal methods can read this appendix selectively: the cross-language table in Section C.2 and the epistemic argument in Section C.8 require no mathematical background. The diagrams in Sections C.3–C.7 are self-contained and preceded by plain-language explanations.

13.1 C.1 The Identification

A **string diagram** is a notation for morphisms in a monoidal category. Wires represent objects (types, states, particles — depending on which language you are using). Boxes represent morphisms (functions, transitions, interactions). Reading left to right is function composition. Wires running in parallel represent a tensor product: two things existing simultaneously and independently.

The result that makes this appendix possible is a theorem, not a conjecture:

> **Coherence Theorem** *(Baez–Lauda, 2011, building on Penrose 1971 and Selinger 2010).* A commutative diagram in a symmetric monoidal category is equal to a string diagram in that category, which is equal to a Feynman

diagram whose vertices are the morphisms. These are three notations for the same mathematical object. A proof in any one notation is a proof in all three simultaneously.

This is not an analogy. It is an identity. A Feynman diagram *is* a morphism in a symmetric monoidal category, drawn as a string diagram. The convergence is not a coincidence of similar-looking notation; it is a theorem about notation.

The Soma-Field Model is defined as a chain of functors between categories (Appendix A.3). Its Lean 4 encoding defines emotions as types and transitions as typed functions between those types (Appendix A.5). By the coherence theorem, both of these are already string diagrams, already Feynman diagrams — they simply have not been drawn as such yet. This appendix draws them.

13.2 C.2 The Cross-Language Table

The following table shows the same mathematical entities in four notations. Nothing is translated between columns. Each row is a single entity viewed through four different naming conventions.

Entity	QFT / Theoretical Physics	Category Theory	Lean 4 Type Theory	Soma-Field Model
Object	Particle state space	Object in category $\mathcal{C}$	Type `α : Type`	Emotional state type
Morphism	Interaction vertex	Morphism $f : A \to B$	Function `f : A → B`	State transition / coupling

Entity	QFT / Theoretical Physics	Category Theory	Lean 4 Type Theory	Soma-Field Model
Identity morphism	Free propagator (no interaction)	$\mathrm{id}_A : A \to A$	`id : α → α`	Emotion persisting unchanged
Composition	Diagram concatenation (left to right)	$g \circ f$	`fun x => g (f x)`	Sequential processing
Tensor product $\otimes$	Two parallel propagators	$A \otimes B$	`A × B` (product type)	Co-occurring emotional states
Symmetry isomorphism	Particle exchange	$\sigma : A \otimes B \cong B \otimes A$	`Prod.swap`	Emotion order-independence
Loop / trace	Closed Feynman loop	Trace in traced monoidal category	Coinductive / recursive type	Rumination; memory feedback
Fixed point	Bound state	Terminal coalgebra of endofunctor	`def` self-referential type	Attractor basin
Vacuum bubble	Zero-order (no-vertex) diagram	Identity on monoidal unit $\mathbf{1}$	`Unit`	Baseline resting state
Propagator	$G(x, y) = \langle \phi(x)\phi(y) \rangle$	Hom-set element	`f : A → B`	Emotional correlation across time

Entity	QFT / Theoretical Physics	Category Theory	Lean 4 Type Theory	Soma-Field Model
Coupling constant	Vertex weight g	Strength of morphism	Function application coefficient	Entry W_{ij} in coupling matrix
Lagrangian	$\mathcal{L}[\phi]$ generating dynamics	Object of $\mathscr{F}$ specifying evolution	`hopfieldEnergy` function	Hopfield energy function $H(\mathbf{e})$
Path integral	$\int e^{iS/\hbar}\,D\phi$	Colimit over morphism paths	`sorry` (requires measure theory)	Langevin stochastic evolution
Spontaneous symmetry breaking	Vacuum choosing a particular minimum	Initial object of attractor category	`inductive AttractorBasin`	Polyvagal state selection

Table C1. Cross-language correspondence. Each row is one mathematical entity in four independent notations. The convergence is not achieved by selecting convenient subsets — every element of the full theory maps. The Hopfield energy function is the Lagrangian. The coupling matrix entries are coupling constants. The Langevin dynamics are a stochastic path integral. The attractor basins are vacua of a spontaneously broken symmetry.

The critical column is the rightmost one. The existence of direct soma-field analogues for every row in this table is not guaranteed by construction — it is a structural fact about the model. A model that required different mathematics in each language would be suspect. A model that requires the same mathematics in all four simultaneously is at least well-formed.

13.3 C.3 The Functor Chain as a String Diagram

The functor chain introduced in Appendix A.3 is already a string diagram. Here it is drawn as one. The reading direction is left to right; boxes are functors (morphisms between categories); thick horizontal lines are the categories (objects) being transformed.

```
READING DIRECTION ──────────────────────────────────────────────────────────►


categories
(wires):    ══ M ══╥══ F ══╥══ S ══╥══ P ══╥══ O ══


functors    ┌────┐ ║ ┌────┐║ ┌────┐║ ┌────┐║
(boxes):    │ L  │─╫─│ Pi │╫─│ M  │╫─│ O  │╫─
            └────┘ ║ └────┘║ └────┘║ └────┘║


plain:      compact  project  apply    render
            ification field   threshold output


            G2 form  int K.E  |e|>T    -> audio
            -> field   dx     -> perc. -> visual
                                        -> MIDI
```

In Lean 4, this is the `pipeline` function from Appendix A.5, composed left to right. In QFT, it is a multi-vertex Feynman diagram with four interaction points. In category theory, it is the composition $O \circ M \circ \Pi \circ \Lambda$. In the Soma-Field Model, it is the full processing chain from M-theory geometry to therapeutic output.

These are not four descriptions of four things that happen to be similar. They are four notations for one thing.

13.4 C.4 Emotional Interaction as a Feynman Vertex

The coupling matrix W specifies how emotional modes interact. Each non-zero entry W_{ij} is a **vertex** in the string diagram — a point at which two emotion wires exchange influence. In QFT, it is a coupling constant. In Lean 4, it is a coefficient in the `hopfieldEnergy` function. In category theory, it is the strength of a morphism.

The simplest illustrative case: two states combining to produce a third. In the context of C-PTSD's asymmetric W, fear and shame frequently do not produce the sum of their individual outputs — they fuse into freeze, a qualitatively different attractor state with lower energy than either component. In string diagram notation:

```
                         W_ij
                  (coupling constant)
                           |
   Fear  ──────────────────[*]────────────────────► Freeze
                         / |
   Shame ─────────────────/  |
                           |
           Lean type: interact : Fear -> Shame -> Freeze
           QFT vertex: two incoming lines, one outgoing
           Soma-field: gradient descent from Fear x Shame
```

```
                                 toward the nearest attractor basin
```

The Feynman rules say: to compute the probability of this interaction, sum over all diagrams with these external lines. In the Soma-Field Model, the analogous computation is the Langevin update step: the field evolves toward lower energy, and the Fear $\otimes$ Shame $\to$ Freeze pathway is the gradient-descent path when W encodes this coupling with sufficient strength.

The Lean 4 expression of this vertex is already in the `hopfieldEnergy` function:

```
-- The coupling W_ij is the vertex weight: influence of mode j on mode i
def coupleStates {n : ⍰} (cm : CouplingMatrix n) (i j : Fin n)
    (e : EmotionalState n) : ⍰ :=
  cm.W i j * e.activation j


-- The full vertex sum over all incoming modes (row i of W·e):
-- This is the Feynman diagram summed over all incoming lines to vertex i
def fieldAtVertex {n : ⍰} (cm : CouplingMatrix n) (e : EmotionalState n)
    (i : Fin n) : ⍰ :=
  Finset.univ.sum (fun j => coupleStates cm i j e)
```

The string diagram, the Feynman vertex, and the Lean function are the same mathematical object. The string diagram makes the topology visible; the Lean function makes it computable; the Feynman rules make it quantitative; the clinical description makes it meaningful.

13.5 C.5 Parallel States: The Tensor Product

When two emotions co-occur — when fear and grief are present simultaneously, neither causing the other — they are not added. They are **tensored**. This is the monoidal product ⊗: two separate wires carrying independent information, processed in parallel. The distinction between the product and sequential composition is categorical, not merely notational.

```
Tensor product (co-occurring, independent):

Fear  ════════════════════════════════════════════════════════► threshold
                    ||
                    || (parallel — both real, both present,
                    ||  neither causes the other)
                    ||
Grief ════════════════════════════════════════════════════════► threshold

vs. sequential composition (causal chain):

Fear ──────────────────►[vertex]──────────────────────────► Grief

(Fear caused Grief: a morphism from one state to another,
 not two states existing simultaneously)
```

This distinction matters clinically. The Soma-Field Model predicts that co-occurring states (tensor product) and causally sequenced states (composition) have different attractor structures, different threshold behaviours, and different therapeutic entry points. They look similar from the outside — the patient presents with fear and grief in either case — but the energy landscape is different, and the gradient-descent path is different.

In Lean 4: `EmotionalState Fear × EmotionalState Grief`. In QFT: two non-interacting propagators running in parallel (their S-matrix is the identity: no vertex connects them). In the energy function: H decouples into $H_{\text{fear}} + H_{\text{grief}}$ when $W_{ij} = 0$ for $i \in \{\text{fear}\}$, $j \in \{\text{grief}\}$.

A therapist who interprets tensored states as composed states — who reads co-occurrence as causation — will construct a different formulation from the one that is actually operating. The string diagram notation makes the topology visible. The categorical distinction between $A \otimes B$ and $A \to B$ is a clinical distinction.

13.6 C.6 The C-PTSD Memory Kernel as a Feynman Loop

In QFT, a **loop diagram** represents a particle that propagates forward, interacts with itself, and propagates onward. Loop corrections are the formal mechanism by which the past influences the present through self-interaction — they are quantum corrections to the classical (loop-free) predictions. A system with no loop diagrams is a classical system; the appearance of loops is the signature of self-referential dynamics.

In Appendix B.2, the C-PTSD memory kernel term is:

$$\int_0^t K(t-s)\,\mathbf{e}(s)\,ds$$

This integral is a loop diagram. The emotional state at past time s propagates forward to present time t, where it enters the current dynamics as an effective self-interaction. The kernel $K(t-s)$ is the propagator of the loop — it determines how strongly the past state at lag $(t-s)$ influences the present.

```
                K(t-s): memory kernel = loop propagator
                (decays with lag: distant past has less weight)
                ┌──────────────────────────────┐
                |                              |
e ──────────────[*]────────────────────────────[*]──────────────► e(t)
        e(s)     ^                              ^
        past     |                              |
        state   vertex 1: past state           vertex 2: loop re-enters
                enters loop                    current dynamics


K = 0 (no trauma):              K != 0 (trauma present):
────────────────────────────►   ────────────[*loop*]────────────────►
clean propagation               propagation with memory self-correction
standard Hopfield               past state alters present gradient
```

In QFT, loop diagrams introduce corrections of order $\hbar$ — the quantum regime absent from the classical (tree-level) theory. In the Soma-Field Model, the memory kernel is analogously a **trauma correction** to the classical Hopfield dynamics: the standard model without K is the tree-level approximation; the C-PTSD modification is the one-loop correction that re-introduces past field configurations into the present gradient.

This is not a metaphor. The mathematics of loop diagrams and the mathematics of non-Markovian memory kernels are formally identical — both are convolution operators over the past trajectory of the field, both produce self-interaction corrections, both vanish in the $K \to 0$ limit, and both create the possibility of dynamics that are qualitatively different from the classical baseline. The QFT analogy is precise because the underlying mathematical structure is the same structure.

13.7 C.7 The Polyvagal Hierarchy as a Phase Diagram

The three polyvagal states — ventral vagal (safe), sympathetic (mobilised), and dorsal vagal (shutdown) — correspond to three distinct local minima of the energy function H. In the string diagram notation for the full system, they are **vacua** of the theory: the states toward which the field naturally evolves under Langevin dynamics, and from which it requires a finite energy input to escape. In QFT, this structure is called spontaneous symmetry breaking — the system must choose one vacuum from among several, and transitions between vacua are non-perturbative.

```
ENERGY LANDSCAPE (schematic H vs. state):


H
^
|          *                    *
|         / \                  / \      (fight/flight: shallow local minima)
|        /   \                /   \
|       /     *──────────────*     \
|      /      (mobilised — saddle)  \
|     /                              \
|    *                                *──────────────────────*
|   (ventral:                         (dorsal:
|    global minimum,                   deep isolated well,
|    regulated calm)                   trapped shutdown)
|
+──────────────────────────────────────────────────────────► state


TRANSITIONS IN STRING DIAGRAM NOTATION:
```

```
┌──────────────────────────────────────────────────────────────────┐
│                                                                  │
│  SAFE / CONNECTED (ventral vagal)                   H = global  │
│  ─────────────────────────────────────────────────  minimum    │
│                        |                                        │
│         "threat signal: |grad H| exceeds T_down"                │
│                        |                                        │
│                        v                                        │
│  FIGHT / FLIGHT (sympathetic)                       H = shallow │
│  ─────────────────────────────────────────────────  local min  │
│                        |                                        │
│         "overwhelm: mobilisation cannot resolve threat"          │
│                        |                                        │
│                        v                                        │
│  FREEZE / SHUTDOWN (dorsal vagal)                   H = deep   │
│  ─────────────────────────────────────────────────  isolated   │
│                                                      well      │
│    <-------- therapeutic re-entry is non-perturbative ---------->│
│    (small perturbations do not escape the well;                 │
│     a qualitatively different intervention is required)         │
│                                                                  │
└──────────────────────────────────────────────────────────────────┘
```

In Lean 4, this is the `AttractorBasin` inductive type with the `classifyAttractor` function (currently `sorry` — the eigenanalysis of the nonlinear system is an open proof obligation). In QFT, it is a theory with three vacua at different energies, where the lowest-energy vacuum (ventral) is the true vacuum and the others are metastable. In category theory, it is the initial object of the attractor sub-category of .

The therapeutic re-entry arrow — from dorsal shutdown toward ventral calm — is the **instanton**. In QFT, an instanton is a non-perturbative process: a trajectory through configuration space that cannot be reached by any sequence of small perturbations, but requires a large, coherent fluctuation. The formal prediction is that moving from deep freeze to regulated calm is not achieved by incremental de-escalation (the gradient flow does not point in that direction from inside the dorsal well), but by a qualitatively different kind of intervention — one that supplies enough energy, in the right direction, to escape the well entirely. This is what somatic therapies describe, and what gradient descent alone cannot provide. The string diagram notation makes the topological reason visible.

13.8 C.8 The Hidden Load-Bearing Assumption

There is a structural tension in the formalism that the string diagram notation exposes directly. It is worth naming explicitly.

The standard Hopfield convergence theorem — the guarantee that the gradient dynamics $\dot{\mathbf{e}} = -\nabla H(\mathbf{e})$ will always reach a fixed point rather than oscillating indefinitely — depends on a single condition: W must be **symmetric** ($W = W^\top$). A symmetric W means that mode i's influence on mode j equals mode j's influence on mode i. Under this condition, H has no saddle-point cycles; the system descends to a minimum and stays there.

Appendix B.2 establishes that the C-PTSD modification breaks this symmetry: the asymmetric component $W_A = \frac{1}{2}(W - W^\top)$ is non-zero, producing limit cycles — persistent oscillations that never reach a fixed minimum. This is not a minor qualification. It is the formal statement that the C-PTSD attractor dynamics are **categorically different** from the standard Hopfield model: not deeper wells, but loops. Not stuck, but cycling.

In string diagram terms, the distinction is topological:

```
Symmetric W (standard Hopfield): all paths are gradient descents

e ──────────────────────────────────────────────────────► attractor
                  (monotone decrease in H; fixed point guaranteed)

Asymmetric W (C-PTSD): some paths are limit cycles

e ────────────────╮
      ^           │
      |           │ (loop: e returns to near-starting point;
      |           │  no fixed point is reached; H oscillates)
      └───────────┘
```

The convergence theorem is the main theorem of the Hopfield network formalism. It is the guarantee that the energy function H does the work we claim it does. For the C-PTSD case, that guarantee does not hold without additional analysis. In the Lean 4 formalization (Appendix A.5), this is the `sorry` inside `classifyAttractor` — but it is a more significant sorry than it first appears, because the asymmetric W case requires a different mathematical tool (Lyapunov stability analysis for non-symmetric systems, or explicit cycle detection) rather than the standard Hopfield argument.

The model is not wrong. The clinical prediction — that C-PTSD produces oscillatory rather than settling dynamics — is the correct prediction, and it follows directly from the asymmetric W. But the Lean formalization as written inherits the standard Hopfield proof strategy in `classifyAttractor`, which does not cover the asymmetric case. That `sorry` is not a placeholder for a routine proof; it is a placeholder for a different proof.

This is precisely the kind of structural gap that the string diagram notation makes visible in a way that prose does not.

13.9 C.9 What This Establishes — and What It Does Not

The cross-language correspondence does not prove the Soma-Field Model is correct. What it establishes is more specific, and more useful.

Structural coherence across independent representational systems. The model is simultaneously legible as a QFT, as a categorical formalism, as a Lean 4 program, and as a clinical description. These four systems were developed independently, with different motivations and different communities. The fact that the same structure appears in all four is evidence that the structure is mathematically natural — that it is, in some sense, the right shape for this problem. A model that required different mathematics in each language would be suspect. A model that requires the same mathematics in all of them is at minimum well-formed: there is nothing incoherent about it that the different languages would independently catch.

The same argument holds in all four languages. When a physical quantity appears identically in classical mechanics, Hamiltonian mechanics, and quantum field theory simultaneously — when the same object emerges in three independent formalisms with the same properties — that convergence is taken as evidence of physical reality. The quantity is not an artifact of any one formalism; it appears in all because it tracks something real. The claim here is structurally analogous: the emotional field $\mathbf{E}(x,t)$, the polyvagal attractor landscape, and the neurodivergent operator modifications appear identically in QFT notation, category theory, Lean 4 types, and clinical description. This does not prove they are physically real. It establishes that they are structurally stable under change of representational language — which is a necessary condition for physical reality, and a condition that models stated only in clinical prose cannot satisfy, because they have only one language.

The gaps are in the same place in all four languages. The two `sorry` markers in Appendix A.5, the asymmetric W gap discussed in Section C.8, and the non-perturbative dorsal-to-ventral transition in Section C.7 are not gaps in one language that happen to be filled in another. They are open questions in all four languages simultaneously. The Lean 4 proof obligation and the QFT instanton calculation and the categorical colimit computation are all versions of the same question: what is the correct formal account of escaping a deep attractor basin? The convergence of gaps is as significant as the convergence of what has been established.

The epistemic asymmetry. The standard biopsychosocial model, as typically stated, has one representational language: clinical prose. It can be questioned, qualified, or dismissed within that language, and there is no independent check. The Soma-Field Model, stated in four languages simultaneously, has a different epistemic structure: to dismiss the model, one must dismiss it in all four languages simultaneously, identifying an error that is present in the QFT formulation, the categorical formulation, the Lean 4 formulation, and the clinical formulation at once. This is a stronger requirement. It does not make the model correct. It makes it harder to dismiss by inattention.

The patient described in Section 1 who was told that persistent somatic symptoms required psychiatric rather than physiological investigation was, among other things, the subject of a single-language model encountering an experience that requires more than one language to describe correctly. A framework expressible in physics, mathematics, formal verification, and clinical description simultaneously does not guarantee the symptoms will be taken seriously. It does ensure that dismissing them requires engaging with the formal structure of the argument, not merely its prose register.

---

14 Appendix D: Perturbative Expansion and Feynman Diagrams

This appendix develops the perturbative sector of the Soma-Field Model — the regime in which emotional modes undergo small fluctuations within an attractor basin. The perturbative expansion organises these fluctuations as a sum over Feynman diagrams, each representing a specific interaction pathway between emotional modes. The non-perturbative sector — threshold crossings and basin transitions — is treated separately as instanton events.

14.1 D.1 The Path Integral and MSR Action

The stochastic Langevin dynamics

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}(t)) + \eta(t)$$

can be reformulated as a path integral using the Martin-Siggia-Rose (MSR) formalism. One introduces a conjugate response field $\tilde{\mathbf{e}}(t)$ and writes:

$$Z = \int D\mathbf{e}\, D\tilde{\mathbf{e}}\ \exp(-S[\mathbf{e}, \tilde{\mathbf{e}}])$$

with the MSR action:

$$S = \int dt\, \tilde{e}_i \Big[\dot{e}_i + \nabla_i H(\mathbf{e})\Big] - \frac{\sigma^2}{2}\, \tilde{e}_i^2$$

Substituting $\nabla_i H = -\sum_j W_{ij} e_j - \theta_i$:

$$S = \int dt\, \tilde{e}_i \Big[\dot{e}_i - \sum_j W_{ij} e_j - \theta_i\Big] - \frac{\sigma^2}{2}\, \tilde{e}_i^2$$

The quadratic part determines the propagator; higher-order terms (from the memory kernel and threshold nonlinearity) give the interaction vertices.

14.2 D.2 The Free Propagator

For a single decoupled mode e_i, the retarded Green's function is:

$$G_i^R(t - t') = \Theta(t - t')\, e^{W_{ii}(t-t')}$$

Θ is the Heaviside step function (causality). For a stable mode, $W_{ii} < 0$ and the propagator decays exponentially at rate $|W_{ii}|$. In Fourier space: $\tilde{G}_i^R(\omega) = (-i\omega - W_{ii})^{-1}$.

```
PROPAGATOR


e_i(t') ──────────────────────────> e_i(t)


        G^R_i(t-t') = Th(t-t') exp(W_ii(t-t'))


        Stable mode (W_ii < 0): decays at rate |W_ii|.
```

```
Unstable mode (W_ii > 0): amplifies until nonlinearity halts it.
```

14.3 D.3 The Coupling Vertex

Off-diagonal couplings W_{ij} $(i \neq j)$ produce the primary interaction vertex:

```
COUPLING VERTEX


    e_i(t) ─────────── e_i'(t)
                 |
             W_ij ●
                 |
    e_j(t) ─────────── e_j'(t)


Excitatory (W_ij > 0): j amplifies i.
Inhibitory (W_ij < 0): j suppresses i.
Asymmetric W_ij =/= W_ji: the interaction is directional.
```

At first perturbative order, the response of mode i to mode j is:

$$e_i(t) \;\supset\; W_{ij} \int_{-\infty}^{t} G_i^R(t-t')\, e_j(t')\, dt'$$

14.4 D.4 Memory Kernel Vertices

The trauma memory kernel introduces a temporally non-local vertex: mode j at time t' drives mode i at time $t > t'$ with weight $K(t-t')$:

```
MEMORY KERNEL VERTEX


e_j(t') ─────┐ K(t-t') J_ij
```

```
        └──────────────── e_i(t)

(integrated over all t' < t)

K(t) = sum_k  A_k exp(-t/tau_k):
each term is a secondary propagator with its own decay rate 1/tau_k.
Multiple kernel terms = multiple memory channels.
```

The memory kernel $K(\tau)$ is itself a sum of propagators. Each traumatic memory channel (A_k, τ_k) is a secondary Green's function: past activations re-enter present dynamics with characteristic timescales τ_k.

14.5 D.5 Feedback Loops and Renormalisation

A closed loop arises when mode i excites mode j which excites mode i again:

```
FEAR-SHAME FEEDBACK LOOP

FEAR  ──────────●─────────────────────● FEAR
                │ W_fs                 │ W_sf
                └──────── SHAME ───────┘

Loop integral: integral dw  G^R_fear(w) W_fs G^R_shame(w) W_sf

Physical consequence: renormalises the effective self-coupling of FEAR
at the loop's characteristic timescale Delta_t ~ 1/(|W_ff| + |W_ss|).
```

Each time the loop cycles, the effective coupling W_{ij} is corrected upward. Repeated co-activation of fear and shame consolidates the coupling — the Feynman diagram account of emotional sensitisation and the development of a conditioned response.

In quantum field theory, loop integrals generate ultraviolet divergences absorbed by renormalisation. The soma-field is regulated by the decay rates $|W_{ii}|$ and τ_k, so the loops are finite. The physical content remains: feedback loops change the effective coupling strength at the loop's timescale. Therapeutic desensitisation is the reversal of this process — damping A_k and reducing $|W_{ij}|$ until the loop no longer closes.

14.6 D.6 Non-Perturbative Sector: Instantons

Threshold crossings — transitions between attractor basins — are invisible to the perturbative expansion. No finite sum of Feynman diagrams describes a topological change of basin. These events are *instantons*: saddle-point solutions of the Wick-rotated action connecting two classical minima.

```
INSTANTON: BASIN TRANSITION (schematic)


H(e)
 |      *                    *
 |     * *     instanton    * *
 |    *   *  ────────────   *   *
 |   *     *______________*     *
 |         A               B
 +──────────────────────────────> e


The instanton is the minimal-action path from basin A to basin B.
Its action S_inst determines the transition rate: Gamma ~ exp(-S_inst).
Deep attractor (large S_inst) = exponentially suppressed escape rate.
Therapeutic intervention increases the effective energy available
to the field, enabling the transition without waiting for spontaneous
```

`tunnelling.`

The non-perturbative amplitude $e^{-S_{\text{inst}}}$ quantifies the difficulty of therapeutic transitions: a deep freeze attractor has a large instanton action and requires sustained, high-energy intervention — not because the model predicts hopelessness, but because it predicts exactly what is needed and why small perturbations will not suffice.

14.7 D.7 Summary: Feynman Rules for the Soma-Field

Element	Mathematical object	Clinical meaning
Propagator $G_i^R(t)$	Green's function of free mode	How a mode evolves between interactions
Coupling vertex W_{ij}	Off-diagonal coupling	One emotion exciting or suppressing another
Memory vertex $K(\tau)J_{ij}$	Non-local temporal vertex	Trauma echo: past drives present
Loop correction	Renormalised coupling	Sensitisation: co-activation strengthens coupling
Instanton	Saddle-point of MSR action	Threshold crossing: basin transition event
External leg	Boundary condition / source	Incoming stimulus or interoceptive cue

The perturbative sector describes the texture within a basin — day-to-day coupling, sensitisation, habituation. The non-perturbative sector describes the pivots. A complete account of emotional dynamics requires both.